Faculty of Engineering and Technology
Electrical and Computer Engineering Department

Project Management
ECON3372

Smart Medical Center Management System for ClinicCare

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1 Project Summary

This project presents the management and execution of the Smart Medical Center Management System for ClinicCare, a web-based system developed to address operational inefficiencies observed during the initial assessment phase. ClinicCare currently relies on manual and paper-based workflows, which resulted in delays in appointment handling, inconsistent patient records, misplaced medical files, and increased administrative workload. These issues were identified during the project initiation phase and defined as the core business problem driving this project.

Based on the approved project charter, the project team planned and managed the design and development of an integrated management system to digitize and automate essential clinic operations. The project scope includes centralized patient registration and medical records management, automated appointment scheduling across departments, digital document archiving, and the integration of an Optical Character Recognition (OCR) module to extract information from scanned medical documents. These components were selected to directly address the identified operational challenges and reduce manual data entry and human error.

From a management perspective, the project was planned over a fixed twelve-week timeline with defined milestones covering requirements analysis, system design, development, testing, deployment, and staff training. Key management decisions included adopting role-based access control to ensure data confidentiality, prioritizing usability for clinic staff, and allocating additional effort to OCR testing due to its technical risk.

The successful delivery of this project is expected to improve workflow efficiency, reduce patient waiting time, and enhance data accessibility within ClinicCare. In addition to delivering the technical system, the project provides a structured management framework and documentation that can support future system expansion and digital transformation initiatives at ClinicCare.

2 Project Initiating

This section presents the initiation phase of the Smart Medical Center Management System for ClinicCare. The following forms document the key outputs required to formally define, authorize, and initiate the project in accordance with standard project management practices.

The checklist below summarizes the required outputs of the project initiation phase. It confirms that all initiation deliverables have been completed and approved before proceeding to the planning phase.

Table 1: Checklist of Project Initiation Outcomes

Checklist: Outcomes of Project Initiation	
Item	**Completed**
Project description (product, process, and/or service)	✓
Feasibility analysis report	✓
Concept document	✓
Project charter	✓
Stakeholder register	✓
Project kickoff meeting	✓

2.1 Project Description

The project description below provides a high-level overview of the proposed system. This description was formulated based on **initial interviews with ClinicCare management** and a **site visit** to observe current manual workflows, ensuring alignment with actual operational needs.

PROJECT DESCRIPTION FORM

Smart Medical Center Management System for ClinicCare

Project Name: Smart Medical Center Management System for ClinicCare
Project Type: Web-Based Information System
Organization: ClinicCare Medical Center

Prepared By: Hala Mohammed (Project Manager) **Date:** December 5, 2025
Reviewed By: Raghad Murad (System Analyst) & Taleen Bayatneh (System Designer) **Date:** December 9, 2025
Approved By: Dr. Ahmad N. Khalil (Clinic Director) **Date:** December 11, 2025

Methodology: This description was derived from initial site observations and stakeholder interviews to ensure the solution addresses real operational bottlenecks.

1. What does the project include?
Design and development of a web-based clinic management system covering patient registration, appointment scheduling, medical document archiving, OCR-based data extraction, and role-based access control.

2. Who is this project for?
ClinicCare administrative staff, doctors, and authorized medical personnel at a small-to-medium medical center operating manual, paper-based workflows.

3. Why is this project important?
ClinicCare currently suffers from delayed services, misplaced medical files, inconsistent patient data, and high administrative workload due to the absence of a centralized digital system. This project eliminates these inefficiencies by digitizing and automating core clinic operations.

4. Expected Value:
Reduced patient waiting time, improved data accuracy, enhanced staff productivity, and a scalable digital foundation for future ClinicCare expansion.

Figure 1: Project Description Form

2.2 Feasibility Analysis Report

FEASIBILITY ANALYSIS REPORT

Smart Medical Center Management System for ClinicCare

Prepared By: Raghad Murad (System Analyst) **Date:** December 15, 2025

Reviewed By: Hala Mohammed (Project Manager) **Date:** December 16, 2025

Methodology: This report evaluates the viability of the proposed system by analyzing current operational inefficiencies, assessing the team's technical capabilities, and comparing deployment alternatives. The analysis is based on data collected during initial site visits and stakeholder interviews.

1. Problem Analysis (Operational Feasibility)

Objective: To determine if the system solves the actual operational issues at ClinicCare.

After analyzing the current manual workflows at ClinicCare, the following critical issues were identified:

- **Inefficiency:** Average patient waiting time exceeds 20 minutes due to manual file retrieval.
- **Data Integrity:** 15% of physical files are misplaced or damaged annually, leading to loss of medical history.
- **Workload:** Administrative staff spend approximately 40% of their time on paper-based data entry rather than patient care.

Note: The above estimates are based on observations during site visits and interviews with clinic staff conducted in December 2025.

Conclusion: The proposed system directly addresses these pain points by digitizing records and automating scheduling. Operationally, the solution is **highly feasible** and necessary.

2. Technical Competencies (Technical Feasibility)

Objective: To assess if the team has the skills and technology to build the system.

The development team has evaluated the technical requirements against current capabilities:

- **Technology Stack:** The team is proficient in modern web development frameworks suitable for building secure and scalable information systems.
- **OCR Integration:** The project plan includes allocating a dedicated developer to integrate established OCR libraries (e.g., Tesseract or Google Vision API).
- **Infrastructure:** ClinicCare possesses the basic hardware (PCs, Internet) required to access the web-based client application.

Conclusion: The project is **technically feasible** with the current team skillset and available technology stack.

3. Solution Alternatives Comparison

Objective: To select the best deployment approach (On-Premise vs. Cloud).

Criteria	Option A: Cloud-Based	Option B: On-Premise
Initial Cost	Low (Subscription/Pay-as-you-go)	High (Server hardware purchase)
Maintenance	Managed by Cloud Provider	Managed by Clinic IT Support
Data Security	High (Enterprise-grade encryption)	Dependent on local IT measures
Accessibility	High (Accessible anywhere)	Limited (Within clinic network)
Recommendation	**Selected for Phase 1**	Future consideration

Final Decision: The project will proceed with a **Cloud-Based Solution**. This decision aligns with the project's limited initial budget and the clinic's lack of dedicated on-site IT maintenance staff.

2.3 Concept Document

PROJECT CONCEPT DOCUMENT

Smart Medical Center Management System for ClinicCare

Prepared By: Taleen Bayatneh (System Designer) **Date:** December 19, 2025

Reviewed By: Hala Mohammed (Project Manager) **Date:** December 21, 2025

Methodology: This document outlines the high-level conceptual solution, defining the strategic goals, the technical approach, and the required resources to establish a clear vision for the development team.

1. Project Goals & Objectives

The primary goal is to transform ClinicCare into a paperless environment:

- **Operational Efficiency:** Reduce patient registration time by 50% through automated workflows.
- **Centralization:** Create a unified repository for patient records accessible by authorized staff.
- **Data Intelligence:** Implement OCR technology to digitize existing paper archives automatically.
- **Scalability:** Design an architecture that supports future expansion to multiple branches.

2. Proposed Technical Approach

- **Frontend:** React.js framework for cross-device compatibility.
- **Backend:** Python (Django Framework) for secure and efficient server-side logic.
- **Database:** PostgreSQL for data integrity and complex medical record queries.
- **Deployment:** Cloud hosting (AWS or Heroku) for high availability and security.

3. Resources & Preliminary Financial Estimates

The budget reflects monthly salaries for a 3-month duration plus infrastructure costs.

Resource Category	Roles Included	Est. Cost (ILS)
Management	Project Manager	10,500
Analysis Team	System Analyst, Requirements Analyst, Business Process Analyst	18,300
Design Team	System Designer, Database Designer, UI/UX Designer	21,600
Development Team	Backend, Frontend, Database, and OCR Developers	36,600
QA & IT Support	Test Engineers, Integration Tester, System Admin	16,800
Non-Human Resources	Software Licenses, Cloud Hosting, APIs	4,000
Subtotal		**107,800**
Contingency (10%)	Risk Reserve	**10,780**
Total Budget		**118,580**

4. Preliminary Risk Assessment

- **Technical Risk:** OCR accuracy may be lower than expected for handwritten notes. *Mitigation: Manual review step for low-confidence scans.*
- **Operational Risk:** Staff resistance to the new system. *Mitigation: Hands-on training sessions.*
- **Security Risk:** Data breaches affecting patient confidentiality. *Mitigation: RBAC and encryption.*

2.4 Project Charter

The project charter formally authorizes the project and defines its objectives, scope, key stakeholders, responsibilities, and major milestones. It serves as a foundational reference for all subsequent project phases.

PROJECT CHARTER

Smart Medical Center Management System for ClinicCare

Prepared By: Hala Mohammed (Project Manager) **Date:** December 25, 2025

Approved By: Dr. Ahmad N. Khalil (Clinic Director) **Date:** January 5, 2026

Methodology: This charter authorizes the project based on the approved feasibility analysis and concept document. It formally commits resources and defines the boundaries for the execution phase.

Figure 2: Project Charter Authorization

2.4.1 General Information

Project Title: Smart Medical Center Management System for ClinicCare

2.4.2 Contact Information

Table 2: Contact Information

Position	Title / Name / Organization	Phone	E-mail
Responsible Executive	Clinic Director – Dr. Ahmad N. Khalil (ClinicCare)	0599-456-321	akhalil@clinicare.ps
Project Manager	Hala Mohammed – Project Team Leader	0598-703-310	hala@bzuedu.ps
Customer Representative(s)	ClinicCare Administration Office	022-984-771	admin@clinicare.ps
Other	IT Support Supervisor – Omar Saeed	0569-332-118	support@clinicare.ps

2.4.3 Project Overview

The Smart Medical Center Management System for ClinicCare is a web-based information system developed to digitize and integrate administrative and operational processes within a medical center. The system replaces manual and paper-based workflows with a secure digital platform that supports patient registration, appointment scheduling, medical document archiving, and administrative monitoring. The solution aims to enhance workflow efficiency, data accessibility, and overall service quality.

2.4.4 Business Problem

ClinicCare currently relies on manual documentation and disconnected processes to manage patient records and appointments. This leads to delayed services, inconsistent or misplaced medical information, increased administrative workload, and difficulty in retrieving patient data. The lack of a centralized digital system negatively affects workflow efficiency, staff productivity, and patient satisfaction.

2.4.5 Project Objectives

- Digitize 100% of new patient registrations and store all patient profiles electronically starting from system go-live.
- Automate appointment scheduling for all clinic departments, enabling at least 90% of bookings to be created/updated via the system (not manual logs) during the first month after deployment.
- Enable secure digital archiving so that 100% of uploaded medical documents are stored with metadata (patient ID, document type, date) and can be retrieved within 10 seconds for 95% of searches.
- Integrate OCR to extract key fields from scanned records with $\geq$95% field-level accuracy on a validation set of at least 100 scanned documents.
- Reduce administrative workload by decreasing manual data entry time by at least 40% compared to the baseline clinic process (measured during pilot week vs. pre-system week).
- Enforce secure, role-based access so that 100% of protected actions require authentication and role permission checks, with 0 high-severity access control findings in security testing.

2.4.6 Project Scope

Product Description:
A web-based medical center management system including patient records management, appointment scheduling, document archiving, OCR-supported data extraction, and administrative monitoring tools.

Acceptance Criteria:

- **Functional Completeness:** All core modules (Patient Registration, Appointments, Medical Records, Document Archive, OCR) pass $\geq$95% of their functional test cases, with 0 critical defects open.
- **Access Control:** Role-based access control is implemented for all user roles, and 100% of restricted actions are blocked for unauthorized roles during testing (0 high-severity access violations).
- **OCR Functionality:** OCR successfully extracts predefined key fields with $\geq$95% field-level accuracy on at least 100 scanned documents, and flags unreadable/low-confidence fields for manual review.
- **Performance & Stability:** The system maintains $\geq$99% uptime during clinic working hours and meets average response time $\leq$2 seconds for 95% of user actions during the pilot.
- **UAT Approval:** Clinic staff approve the system after UAT, with $\geq$90% of UAT scenarios passed and an overall satisfaction score of $\geq$80%.

Deliverables:

- Fully functional clinic management system
- OCR document processing module
- Configured database and hosting environment
- Testing and quality assurance report
- User manual and technical documentation

Exclusions:
Billing systems, insurance processing, telemedicine services, mobile applications, and external hospital integrations.

Constraints:
Fixed 12-week timeline, limited development team resources, and clinic staff availability.

Assumptions:
ClinicCare will provide timely feedback and required technical infrastructure.

2.4.7 Major Milestones / Critical Success Factors

Table 3: Major Project Milestones

Milestone	Deadline
Requirements Gathering and Approval	Week 2
System Design Completion	Week 5
Core System Development	Week 8
System Testing and Validation	Week 11
Deployment, Training, and Handover	Week 12

2.4.8 Authority and Responsibility

Authorization (Executive Authorizing the Project):
ClinicCare Management formally authorizes this project and approves its objectives, scope, schedule, and major deliverables.

Project Manager (Authority and Responsibility):
The Project Manager is responsible for planning, coordination, communication, progress monitoring, risk management, and ensuring that project deliverables meet approved requirements within the defined timeline.

2.4.9 Roles and Responsibilities of Other Stakeholders

ClinicCare Administration provides operational requirements and validates workflows. Doctors and medical staff review system usability and patient record handling. Reception staff validate appointment scheduling processes. IT support assists with hosting, system maintenance, and access control.

2.4.10 Project Organization

The project organization consists of ClinicCare Management as the executive sponsor, a Project Manager overseeing planning and execution, and a development team responsible for system design, implementation, testing, and deployment.

2.4.11 Disaster Recovery Methodology

The system will implement regular automated database backups and secure storage of uploaded medical documents. Backup copies will be stored in a separate secure location. Recovery procedures will enable restoration of system functionality in case of system failure, data corruption, or accidental data loss.

2.4.12 Resources and Funding

Table 4: Resources and Funding

Resource Category	Source / Description	Funding / Value (ILS)
Human Resources	Project Team (Management, Analysis, Design, Dev, QA)	107,800
Facilities	ClinicCare Labs	In-Kind (0)
Equipment	Development PCs	In-Kind (0)
Software	Open-source Tools, IDEs	Minimal Cost (500)
Materials	Documentation Resources	Minimal Cost (200)
Non-Human (Tech)	Cloud Hosting, APIs, and Infrastructure	3,300
Contingency Reserve	Risk Management Reserve (10%)	11,180
Total Budget	**Authorized by ClinicCare Management**	**122,980**

2.4.13 Signatures

Table 5: Authorization Signatures

Position / Title	Signature	Date
Responsible Executive	.....	27/12/2025
Project Manager	.....	27/12/2025
Other Stakeholders (as needed)	.....	27/12/2025

2.5 Stakeholder Register

The stakeholder register identifies all individuals and groups that may affect or be affected by the project. Each entry documents the stakeholder's role, level of influence, and potential impact on project success.

STAKEHOLDER REGISTER

Smart Medical Center Management System for ClinicCare

Prepared By: Raghad Murad (System Analyst) **Date:** December 27, 2025

Reviewed By: Hala Mohammed (Project Manager) **Date:** December 30, 2025

Methodology: Stakeholders were identified through organizational charts analysis and interviews with clinic management to ensure no critical party is overlooked.

Figure 3: Stakeholder Register Overview

2.5.1 Stakeholder 1: Dr. Ahmad N. Khalil

Stakeholder Name: Dr. Ahmad N. Khalil
Position: Clinic Director – ClinicCare (Project Sponsor)

Contact Information: Phone: 0599-456-321 — E-mail: akhalil@clinicare.ps

Potential Impact on Project:
As the project sponsor and clinic director, Dr. Ahmad Khalil has a high level of authority and influence over the project. He is responsible for approving the project charter, scope, budget, and major decisions. His support directly affects resource availability, priority setting, and final system approval.

Miscellaneous Comments:
Requires high-level progress reports and milestone updates. Strong supporter of digital transformation initiatives.

2.5.2 Stakeholder 2: Hala Mohammed

Stakeholder Name: Hala Mohammed
Position: Project Manager / Project Team Leader

Contact Information: Phone: 0598-703-310 — E-mail: hala.mohammed@bzuedu.ps

Potential Impact on Project:
The Project Manager has direct responsibility for planning, execution, monitoring, and communication. This role significantly impacts project success through schedule control, risk management, stakeholder coordination, and quality assurance.

Miscellaneous Comments:
Acts as the main point of contact for all stakeholders and is responsible for issue resolution and reporting.

2.5.3 Stakeholder 3: Ms. Lina Haddad

Stakeholder Name: Ms. Lina Haddad
Position: Administrative Manager – ClinicCare

Contact Information: Phone: 022-984-771 — E-mail: lina.haddad@clinicare.ps

Potential Impact on Project:
As a key operational user, the administrative manager influences system requirements, workflow design, and daily usage. Her feedback is essential to ensure that the system supports appointment scheduling, patient registration, and document handling efficiently.

Miscellaneous Comments:
Requires early training and continuous involvement to ensure smooth system adoption.

2.5.4 Stakeholder 4: Dr. Samer Yaseen

Stakeholder Name: Dr. Samer Yaseen
Position: Senior Physician – ClinicCare

Contact Information: Phone: 0592-771-884 — E-mail: samer.yaseen@clinicare.ps

Potential Impact on Project:
Doctors rely on accurate, up-to-date patient records for diagnosis and treatment. System usability and data accuracy directly affect their efficiency and acceptance of the system.

Miscellaneous Comments:
Interested in fast access to patient history and minimal disruption to clinical workflows.

2.5.5 Stakeholder 5: Reception Staff Team

Stakeholder Name: Reception Staff Team – ClinicCare
Position: Front Desk and Appointment Scheduling Staff

Contact Information: Phone: 022-984-780 — E-mail: reception@clinicare.ps

Potential Impact on Project:
Reception staff are daily users of the appointment scheduling and patient registration modules. Their efficiency directly affects patient waiting time and service quality.

Miscellaneous Comments:
Requires intuitive interface design and hands-on training sessions.

2.5.6 Stakeholder 6: Omar Saeed

Stakeholder Name: Omar Saeed
Position: IT Support Supervisor – ClinicCare

Contact Information: Phone: 0569-332-118 — E-mail: it.support@clinicare.ps

Potential Impact on Project:
IT support is responsible for system hosting, maintenance, backups, and access control. Their involvement is critical for system stability, security, and disaster recovery.

Miscellaneous Comments:
Requires technical documentation and clear handover procedures.

2.5.7 Stakeholder 7: ClinicCare Patients

Stakeholder Name: ClinicCare Patients
Position: Indirect Stakeholders / Service Beneficiaries

Contact Information: General Clinic Contact: 022-984-700 — E-mail: info@clinicare.ps

Potential Impact on Project:
Patients benefit from improved appointment scheduling, reduced waiting times, and better management of medical records. While they have low direct influence, their satisfaction reflects project success.

Miscellaneous Comments:
System improvements should prioritize service efficiency and data confidentiality.

2.5.8 Stakeholder 8: Project Development Team

Stakeholder Name: Project Development Team (Internal)
Position: System Designers, Backend/Frontend Developers, QA Engineers, and IT Support

Contact Information: Managed via Project Manager

Potential Impact on Project:
This group is responsible for the technical execution, coding, testing, and deployment of the system. Their technical skills, availability, and morale directly impact the project's timeline and quality.

Miscellaneous Comments:
Detailed roles and individual responsibilities are documented in the *Human Resource Plan* and *Project Team Directory*.

2.6 Kickoff Meeting Agenda

The kickoff meeting agenda outlines the structure and objectives of the initial project meeting. It ensures alignment among stakeholders and marks the formal transition from the initiation phase to the planning phase.

MEETING AGENDA

PROJECT KICKOFF: CLINICCARE SMART MANAGEMENT SYSTEM

Organized By: Hala Mohammed (Project Manager)

Date: January 5, 2026

Time: 9:00 AM

Location: ClinicCare Main Branch – Conference Room

Attendees: Project Team, ClinicCare Management, and Key Stakeholders

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AGENDA ITEMS:

1. **Welcome and Introduction**
   a. Opening remarks by the Project Manager.
   b. Overview of meeting objectives.
2. **Team Introductions**
   a. Introduction of the project team members.
   b. Introduction of ClinicCare management and staff.
3. **Project Charter Review**
   a. Project overview, background, and business problem.
   b. Discussion of assumptions, constraints, and risks.
   c. Approval of project scope, milestones, and deliverables.
   d. Confirmation of roles, responsibilities, and authority.
   e. Review of resources, budget, and disaster recovery plan.
4. **Stakeholder Communication Plan**
   a. Review of identified stakeholders.
   b. Agreement on communication channels and frequency.
5. **Next Steps**
   a. Formal transition to the Planning Phase.
   b. Assignment of initial WBS and scheduling tasks.
   c. Schedule for weekly status meetings.
6. **Q&A Session**
   Open floor for questions, concerns, and feedback.

MEETING OUTCOME:

All participants confirmed their understanding of the project objectives and scope. The project charter was formally approved, and the team is authorized to proceed with the Planning Phase effective immediately.

Figure 4: Kickoff Meeting Agenda

3 Project Planning

This section presents the planning phase of the Smart Medical Center Management System for ClinicCare. The planning phase defines how the project objectives will be achieved through detailed scheduling, cost estimation, resource allocation, communication, quality assurance, procurement, and risk management.

3.1 Project Schedule Planning

This subsection defines the detailed project schedule, including task decomposition, sequencing, duration estimation, and milestone planning.

3.1.1 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) decomposes the ClinicCare project into manageable work packages to support scheduling, cost estimation, and responsibility assignment.

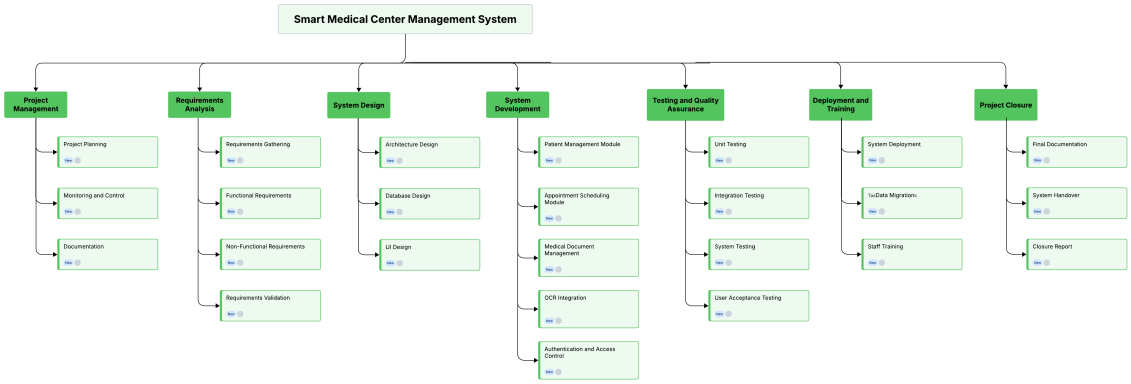

Figure 5: Work Breakdown Structure (WBS) for the ClinicCare Project

3.1.2 WBS Dictionary

Table 6: WBS Dictionary for ClinicCare Project

WBS ID	Task Name	Description	Responsible
1.0	Project Management	Overall planning, coordination, monitoring, and control of the project	Project Manager
2.1	Requirements Gathering	Collecting requirements from clinic administration, doctors, and staff	Project Manager
3.2	Database Design	Designing database schema for patients, appointments, and documents	System Designer
4.1	Patient Management Module	Development of patient registration and medical record management features	Development Team
4.4	OCR Integration Module	Integration of OCR technology for extracting data from scanned documents	Development Team
5.4	User Acceptance Testing	Validation of system functionality by clinic staff	ClinicCare Users
6.3	Staff Training	Training clinic staff on system usage and workflows	Project Manager
7.2	System Handover	Formal delivery of system and documentation to ClinicCare	Project Manager

3.1.3 Activity List

Table 7: ClinicCare Project Activity List

Activity ID	Activity Name	Predecessor(s)	Responsible Role
A1	Requirements Gathering and Analysis	–	Project Manager / System Analyst
A2	System Architecture and UML Design	A1	System Designer
A3	Database Design and Setup	A2	Development Team
A4	Patient Records Module Development	A3	Backend Developer
A5	Appointment Scheduling Module Development	A3	Backend Developer
A6	OCR Integration for Document Scanning	A4	OCR Specialist
A7	Admin Dashboard and Access Control Implementation	A4	Development Team
A8	System Integration and Testing	A5, A6, A7	QA Team
A9	User Acceptance Testing (UAT)	A8	ClinicCare Users
A10	Deployment, Training, and Go-Live	A9	Project Manager / IT Support

3.1.4 Activity Sequencing and CPM Network Diagram

This subsection illustrates the logical relationships and dependencies among project activities using a Critical Path Method (CPM) network diagram.

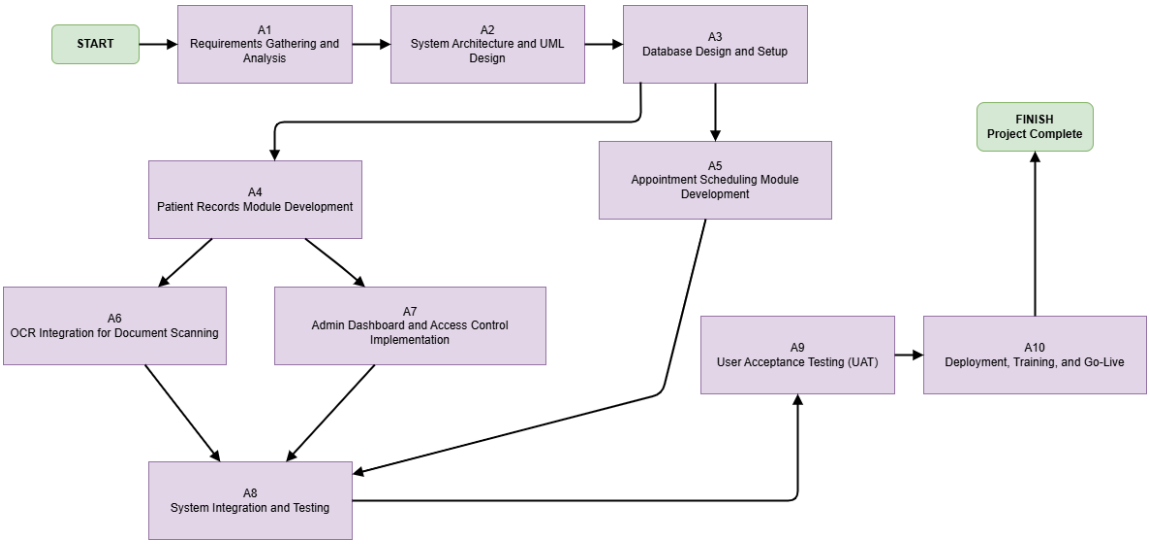

Figure 6: CPM Network Diagram for the ClinicCare Project

3.1.5 Project Schedule Development

This subsection presents the finalized project schedule in the form of a Gantt chart to visualize activity timing and overall project timeline.

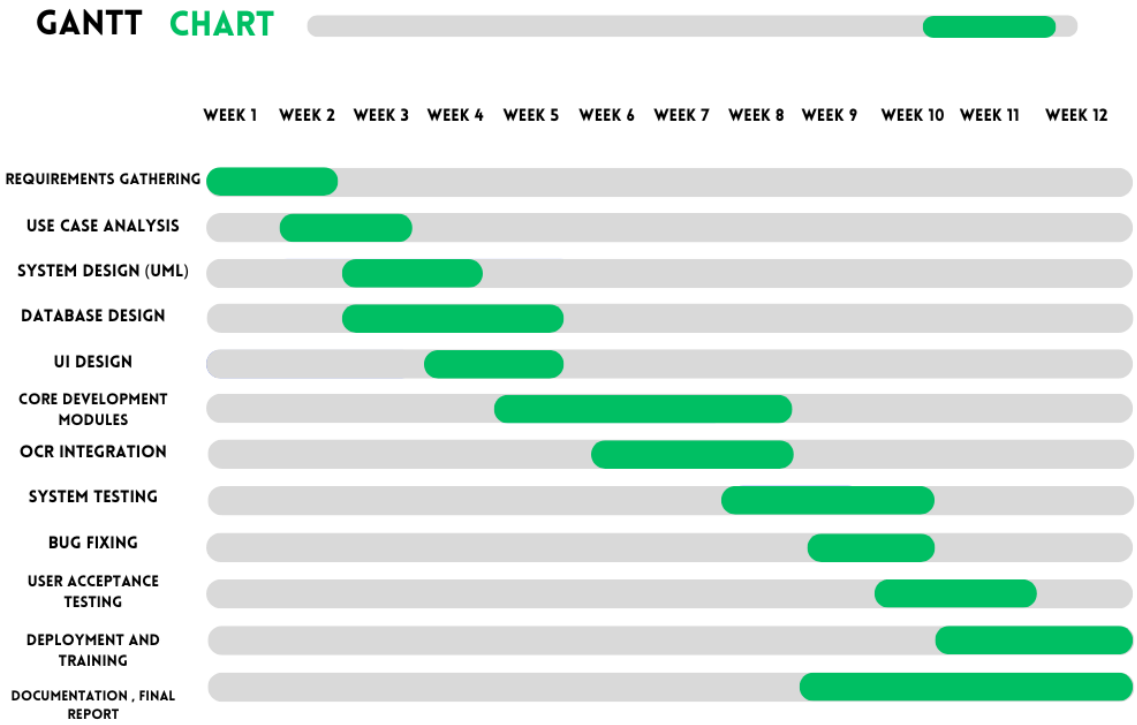

Figure 7: ClinicCare Project Gantt Chart (12-Week Schedule)

3.1.6 Project Milestones

Table 8: Project Milestones

Milestone	Planned Date	Responsible
Requirements Approval	End of Week 2	Project Manager
System Design Completed	End of Week 4	System Designer
Core Development Completed	End of Week 8	Development Team
System Testing Completed	End of Week 10	QA Team
Deployment and Handover	End of Week 12	Project Manager

3.2 Project Cost Planning

3.2.1 Cost Estimation Framework

The following cost categories were considered during the cost estimation process for the ClinicCare project, based on standard project management practices.

Cost Categories Considered:

- Labor (personnel)
- Materials
- Equipment and technology
- Services
- Facilities
- Cost of money
- Inflation allowance
- Contingency
- Risk
- Overhead (indirect costs)

Figure 8: Cost Categories Considered for Project Cost Estimates

3.2.2 Resources Used for Cost Estimation

The following resources were utilized to develop accurate cost estimates for the ClinicCare project.

Essential Resources:

- Project scope statement with WBS and WBS dictionary
- Project schedule
- Human resource plan
- Risk register
- Enterprise environmental factors

Figure 9: Essential Resources for Developing Cost Estimates

3.2.3 Project Estimation Form

PROJECT ESTIMATION FORM

ClinicCare – Smart Medical Center Management System

Prepared By: Project Management Team **Date:** March 2026
Approved By: ClinicCare Management **Estimate No.:** CE-PM-02

Deliverable / Component	Qty	Material (ILS)	Equipment (ILS)	Labor Hours	Cost (ILS)
Requirements Analysis	1	0	0	160	4,800
System Architecture Design	1	0	0	140	4,200
Database Design	1	0	0	130	3,900
Patient Management Module	1	0	1,200	240	8,400
Appointment Scheduling Module	1	0	1,000	200	7,000
Medical Document Management	1	0	1,000	180	6,000
OCR Integration Module	1	600	1,800	220	9,600
System Testing & QA	1	0	800	160	5,600
Deployment & Staff Training	1	400	1,000	120	4,400
				Subtotals	**53,900**
				Plus Indirect Costs	**12,100**
				Total Estimated Project Cost (Development)	**66,000**

Table 9: Project Cost Estimation Summary Form

Comments:

The revised cost estimates reflect a more realistic commercial-scale development effort, including increased labor hours, specialized OCR integration costs, testing effort, and risk contingency allowances based on project complexity.

Note: This project estimation represents the **cost performance baseline** for development work only. It includes direct costs and indirect costs (overhead). The **authorized project budget** is presented in Table 9 and includes additional project management and quality oversight activities, as well as contingency reserves.

3.2.4 Project Budget Summary Form

Project Name: Smart Medical Center Management System (ClinicCare)					
Project Manager: Project Management Team					
ID	**Project Task**	**Hours**	**Labor (ILS)**	**Other (ILS)**	**Total (ILS)**
1	**Human Resources (3-Month Salaries)**				
1.1	Management (Project Manager)	350	10,500	0	10,500
1.2	Analysis Team	277	8,300	0	8,300
1.3	Design Team	320	9,600	0	9,600
1.4	Development Team (incl. OCR)	2,200	66,000	0	66,000
1.5	QA & IT Support	314	9,400	0	9,400
	Subtotal (Human Resources)	**3,461**	**103,800**	**0**	**103,800**
2	**Non-Human Resources (Infrastructure)**				
2.1	Software Licenses, Cloud Hosting, APIs	0	0	4,000	4,000
	Subtotal (Non-Human Resources)	**0**	**0**	**4,000**	**4,000**
				Sub-total	**107,800**
				Contingency (10%)	**10,780**
				TOTAL PROJECT BUDGET	**118,580**

Table 10: Project Budget Summary Form for ClinicCare

3.3 Human Resource Planning

3.3.1 Roles and Responsibilities

This subsection defines the roles required for the successful execution of the ClinicCare project. Each role is documented in a standardized form outlining its authority, responsibility, and required competency.

Project Manager

ROLE: Project Manager

AUTHORITY:
The Project Manager has full authority to plan, coordinate, and control project activities, approve schedules, allocate resources, and communicate with stakeholders.

RESPONSIBILITY:
Responsible for overall project planning, monitoring progress, managing risks, coordinating the project team, ensuring deliverables meet scope, time, cost, and quality requirements.

COMPETENCY:
Strong leadership, project management skills, knowledge of software development processes, and experience with planning and reporting tools.

Figure 10: Project Manager Role Description

System Analyst

ROLE: System Analyst

AUTHORITY:
Authorized to define, document, and validate system requirements in coordination with clinic stakeholders and the project manager.

RESPONSIBILITY:
Responsible for requirements gathering, documentation of functional and non-functional requirements, and validation before design and development.

COMPETENCY:
Expertise in system analysis, requirement elicitation, documentation, and communication with technical and non-technical stakeholders.

Figure 11: System Analyst Role Description

System Designer / Architect

ROLE: System Designer / Architect

AUTHORITY:
Authorized to define system architecture, database design, and major technical decisions aligned with approved requirements.

RESPONSIBILITY:
Responsible for system architecture design, database schema design, UI structure, and ensuring scalability, security, and performance.

COMPETENCY:
Strong technical background in software architecture, database design, and system modeling techniques.

Figure 12: System Designer Role Description

Software Development Team

ROLE: Software Development Team

AUTHORITY:
Authorized to implement system components based on approved designs and suggest technical improvements.

RESPONSIBILITY:
Responsible for coding, integration, and unit testing of system modules including patient management, appointment scheduling, OCR integration, and access control.

COMPETENCY:
Proficiency in web development technologies, database integration, and secure coding practices.

Figure 13: Development Team Role Description

IT Support / Technical Support

ROLE: IT Support / Technical Support

AUTHORITY:
IT Support personnel have limited decision-making authority and do not control project budget or scope. However, they are authorized to perform system maintenance tasks, troubleshoot technical issues, and escalate critical problems to the Project Manager when required.

RESPONSIBILITY:
Responsible for providing technical support during system deployment and operation, handling user-reported issues, performing routine system maintenance, assisting with system configuration, and supporting users during the transition and early usage phases of the ClinicCare system.

COMPETENCY:
Must have technical knowledge of the deployed system, basic networking and database concepts, troubleshooting skills, and the ability to communicate effectively with non-technical users.

Figure 14: IT Support Role Description

Quality Assurance Team

ROLE: Quality Assurance Team

AUTHORITY:
Authorized to evaluate system quality, report defects, and recommend corrective actions prior to system acceptance.

RESPONSIBILITY:
Responsible for test planning, system testing, integration testing, and support of user acceptance testing.

COMPETENCY:
Experience in software testing methodologies, defect tracking, and test documentation.

Figure 15: Quality Assurance Role Description

ClinicCare Users / Customer Representatives

ROLE: ClinicCare Users / Customer Representatives

AUTHORITY:
Authorized to approve system functionality during user acceptance testing and provide operational feedback.

RESPONSIBILITY:
Responsible for validating system usability, participating in acceptance testing, and confirming that system requirements are met.

COMPETENCY:
Operational knowledge of clinic workflows and familiarity with system usage.

Figure 16: ClinicCare User Role Description

3.3.2 Responsibility Assignment Matrix (RACI)

The following RACI chart summarizes the roles and responsibilities of project team members across major project activities.

Activity	Project Manager	System Analyst	System Designer	Dev Team	QA Team	IT Support
Requirements Gathering	A	R	C	I	I	I
System Design	A	C	R	I	I	I
Database & Architecture Design	A	I	R	C	I	I
System Development	A	I	C	R	I	I
OCR Integration	A	I	C	R	I	I
System Testing	A	I	I	C	R	I
User Acceptance Testing	A	C	I	I	R	I
Deployment	A	I	I	C	I	R
Staff Training	A	I	I	I	C	R
Project Monitoring & Reporting	R	I	I	I	I	I

Table 11: RACI Chart for ClinicCare Project

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

3.3.3 Organization Chart

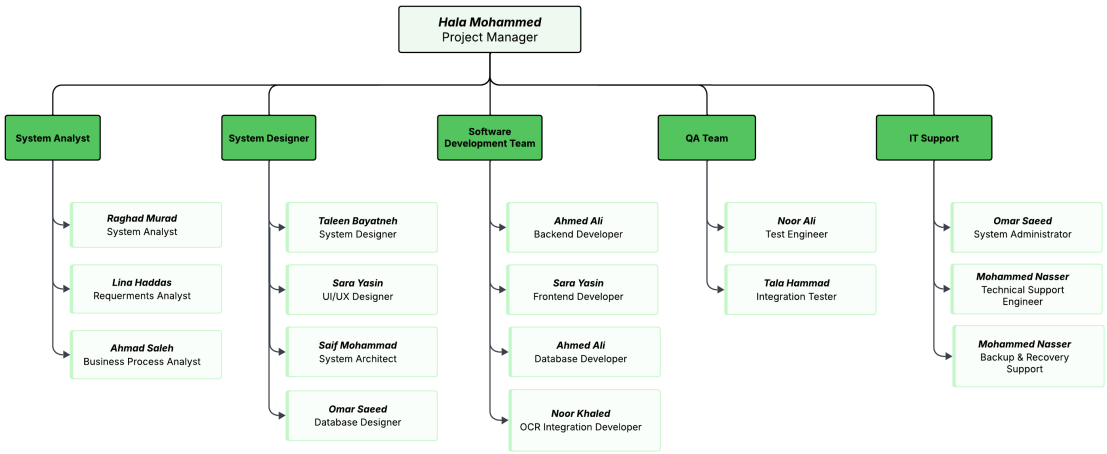

Figure 17: Hierarchical Team Organization Chart

3.3.4 Staffing

The staffing plan defines how and when human resources will be acquired, utilized, and released throughout the project lifecycle. It ensures that the right skills are available at the right time while maintaining cost efficiency and avoiding resource over-allocation.

Staff Acquisition

All human resources required for the ClinicCare project are internally assigned. Team members are selected based on their technical expertise and availability within the project team. No external hiring, contractors, or procurement processes are required. The project manager is authorized to assign roles and responsibilities directly, ensuring flexibility and rapid decision-making.

Staffing Time Frames and Resource Allocation

This element summarizes the workload distribution of project team members throughout the 12-week project duration. Resource requirements are expressed in working hours rather than headcount to provide flexibility in scheduling and cost control.

Values represent average daily working hours per week, where **8** indicates full-time engagement, **4** indicates part-time or support involvement, and "–" indicates periods of no assignment.

Resource	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Management & Analysis												
Hala Mohammed (PM)	8	8	8	8	8	8	8	8	8	8	8	8
Raghad Murad	8	8	4	–	–	–	–	–	–	–	–	–
Lina Haddad	8	8	–	–	–	–	–	–	–	–	4	4
Ahmad Saleh	8	8	–	–	–	–	–	–	–	–	–	–
System Design												
Taleen Bayatneh	–	–	8	8	4	–	–	–	–	–	–	–
Saif Mohammad	–	–	8	4	–	–	–	–	–	–	–	–
Omar Saeed (DB)	–	–	8	8	–	–	–	–	–	–	–	4
Sara Yasin (UI)	–	–	8	8	8	4	4	–	–	–	–	–
Development												
Ahmed Ali (Back)	–	–	–	8	8	8	8	8	4	–	–	–
Sara Yasin (Front)	–	–	–	8	8	8	8	8	4	–	–	–
Ahmed Ali (DB Dev)	–	–	–	8	8	–	–	–	–	–	–	–
Noor Khaled (OCR)	–	–	–	–	–	8	8	8	–	–	–	–
QA & Testing												
Noor Ali (QA)	–	–	–	–	–	–	–	–	8	8	8	4
Tala Hammad (Int.)	–	–	–	–	–	–	–	–	–	8	8	–
IT Support												
Omar Saeed (Admin)	–	–	–	–	–	–	–	–	–	–	8	8
Mohd. Nasser (Tech)	–	–	–	–	–	–	–	–	–	–	8	8

Figure 18: Overall Staffing Time Frames Across the 12-Week Project Timeline

Figure 18 illustrates the phased allocation of human resources across the project life-cycle. Management and analysis activities are concentrated in the early weeks, system design activities dominate the mid-phase, development resources peak during the execution phase, while quality assurance, deployment, and IT support activities are scheduled toward the final weeks. This phased allocation minimizes idle time and optimizes resource utilization.

Figure 19: Individual Resource Utilization and Workload Intensity

Figure 20: Individual Resource Utilization and Workload Intensity

Figures 19 and 20 provide a detailed view of individual team member engagement and workload intensity. These charts support capacity planning by ensuring that no team member is over-allocated and that specialized roles are engaged only during the phases where their expertise is required.

Team-Based Time Frame Histograms

To further clarify workload distribution, the following histograms present the time frames for each major functional team involved in the project.

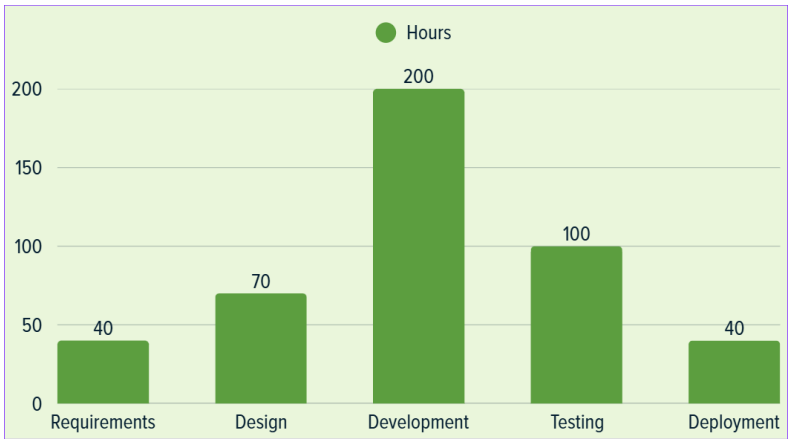

Figure 21: Time Frame Histogram for Software Development Team

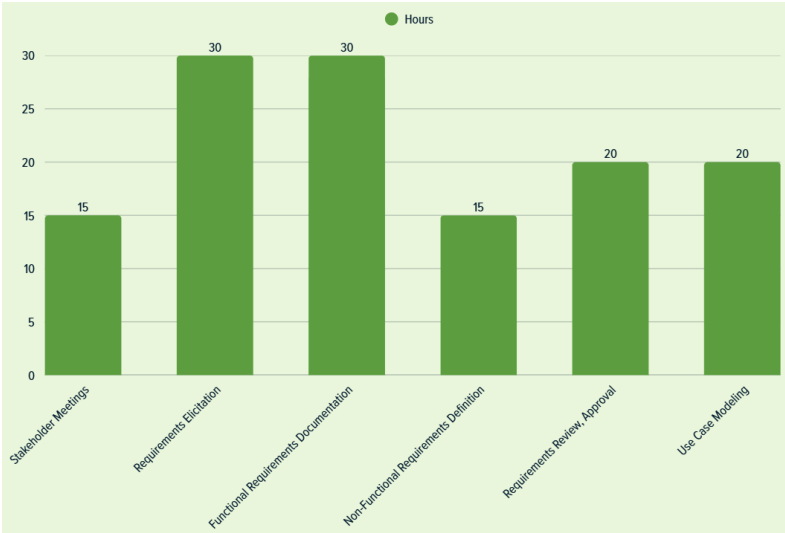

Figure 22: Time Frame Histogram for Requirements Analysis Team

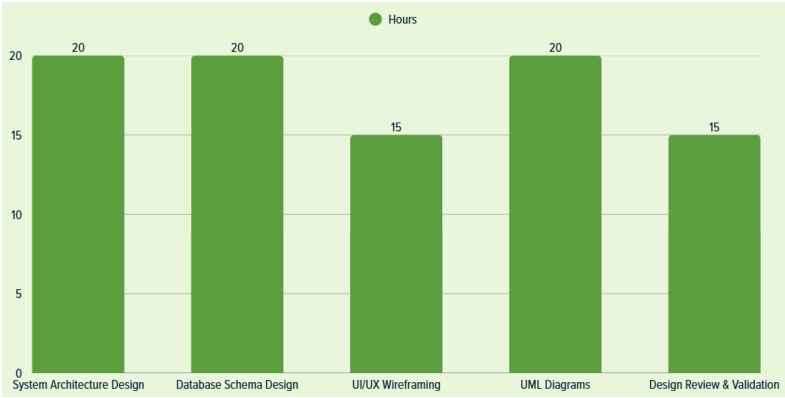

Figure 23: Time Frame Histogram for System Design Team

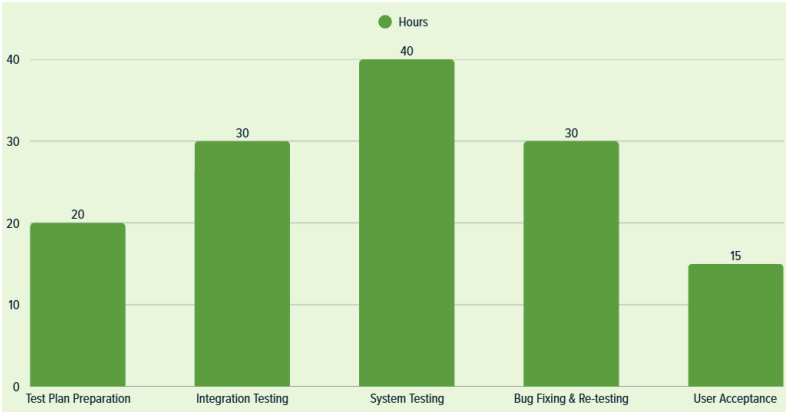

Figure 24: Time Frame Histogram for Testing and Quality Assurance Team

Figure 25: Time Frame Histogram for Deployment and Training Team

These team-based histograms confirm that resource utilization aligns with project phases and deliverables, ensuring effective coordination between teams and smooth handoffs between analysis, design, development, testing, and deployment activities.

Staff Release Method

Team members are formally released from the project upon completion of their assigned deliverables. Analysts and designers are released following the completion of the design phase, while development, QA, and IT support staff remain engaged until successful system deployment, acceptance, and knowledge transfer. Formal staff release occurs during project closure to ensure that all work is completed, documented, and financially accounted for.

3.3.5 Training Needs

TRAINING PLAN

Target Participants: Clinic administrative staff, reception staff, medical staff, IT support

Training Content: System usage, appointment scheduling, document management, OCR usage, security and access control

Training Period: Weeks 11–12

Responsible Party: Project Manager and IT Support

Figure 26: Training Needs and Training Plan

3.3.6 Recognition and Rewards

Recognition Type	Description
Formal Acknowledgment	Recognition during meetings and final presentation
Milestone Completion	Documentation of individual contributions
Professional Growth	Skill development and experience gained

Table 12: Recognition and Reward Mechanisms

3.3.7 Compliance

All project members must comply with confidentiality policies and ethical guidelines. Role-based access control is enforced to protect sensitive medical data and ensure compliance with organizational standards.

Figure 27: Compliance Guidelines

3.3.8 Safety

Safety considerations focus on data protection and operational reliability. Secure authentication, controlled access privileges, and regular backups are implemented to minimize risks and ensure system stability.

Figure 28: Data and Operational Safety Measures

3.4 Communication Management Plan

The purpose of this plan is to ensure that all project stakeholders receive the right information at the right time, and in an appropriate format throughout the project lifecycle. Effective communication is critical to project success, as it supports decision making, risk management, coordination between teams, and stakeholder engagement.
The communication plan identifies stakeholder communication requirements, information types, communication frequency, responsible parties, and communication channels. It also establishes a clear chain of command for issue escalation and ensures that confidential medical and project information is shared securely and appropriately.

3.4.1 Communication Plan With Stakeholders

Stakeholder	Information Type	Frequency	Communication Channel
Project Manager	Detailed progress, issues, risks	Weekly	Team meetings, email
Clinic Administration	Requirements updates, workflow changes	Weekly	Meetings, email
Doctors and Medical Staff	System functionality, usability feedback	As needed	Meetings, feedback forms
Reception Staff	Appointment module updates, training info	As needed	Training sessions, email
Development Team	Tasks, technical issues, deadlines	Daily / Weekly	Meetings, messaging tools
QA Team	Test plans, defects, fixes	Weekly	Reports, issue tracking system
IT Support	Deployment plans, technical documentation	As needed	Technical documentation, meetings

Table 13: Communication Matrix for ClinicCare System

3.4.2 Communication Methods

- Formal meetings
- Progress reports (weekly or monthly)
- Email communication

- Online collaboration tools for internal team coordination
- System demonstrations and training sessions for clinic staff

3.4.3 Communication Authority and Confidentiality

COMMUNICATION AUTHORITY AND CONFIDENTIALITY

Purpose of Communication:
Ensure timely information exchange, stakeholder alignment, effective issue resolution, and secure handling of medical and project-related information.

Authority to Approve Confidential Information:

- Project Manager – Approval of internal project communications
- Clinic Director / Project Sponsor – Approval of sensitive medical and organizational information

Authorized Recipients of Confidential Information:
Project Manager, Clinic Director, authorized medical staff, and IT Support Supervisor (technical and security-related data only).

Figure 29: Communication Authority and Confidentiality

3.4.4 Communication Chain of Command

COMMUNICATION ESCALATION FLOW

Project Team Member
↓
Project Manager
↓
Clinic Director / Project Sponsor

Figure 30: Communication Chain of Command

3.4.5 Communication Constraints

COMMUNICATION CONSTRAINTS

- Medical data confidentiality and patient privacy regulations
- Organizational security and access control policies
- Limited availability of clinic staff during operational hours
- Restricted access to sensitive patient and system information

Figure 31: Communication Constraints

3.4.6 Communication Flow and Information Exchange

PROJECT COMMUNICATION FLOW

- Project Team → Project Manager (task updates, issues, risks)
- Project Manager → ClinicCare Management (progress reports, approvals)
- Clinic Staff → Project Manager (feedback, operational requirements)
- Project Manager → Development Team (clarifications, change requests)

Figure 32: Project Communication Flow

3.4.7 Communication Resources

Table 14: Communication Resources Allocation

Resource	Purpose	Frequency
Team Meetings	Coordination and progress tracking	Weekly
Progress Reports	Project status and management updates	Weekly
Email Communication	Formal notifications and documentation	As needed
Online Collaboration Tools	Internal coordination and task tracking	Continuous
Training Sessions	Knowledge transfer and system usage	Final Phase

3.4.8 Communication Plan Update Method

COMMUNICATION PLAN UPDATE METHOD

The communication management plan will be reviewed and updated by the Project Manager at major project milestones or when changes occur in stakeholder involvement, project scope, or communication effectiveness.

Figure 33: Communication Plan Update Method

3.4.9 Communication Templates and Guidelines

Table 15: Communication Templates and Guidelines

Template Type	Description
Meeting Agenda Template	Standard format for project meetings
Meeting Minutes Template	Documentation of decisions and action items
Progress Report Template	Weekly project status reporting
Email Communication Guidelines	Formal structure and confidentiality rules

3.4.10 Communication Glossary

Table 16: Communication Glossary

Term	Definition
Stakeholder	Individual or group affected by the project
Milestone	Significant project event or deliverable
Escalation	Transfer of an issue to higher authority
Confidential Information	Restricted medical or project-related data

3.5 Procurement Management Plan

Although the ClinicCare project is primarily developed internally by the project team, certain external resources and services are required to ensure successful completion. Procurement planning is conducted to identify these resources, evaluate potential suppliers, and coordinate delivery with the project schedule.

3.5.1 Resources to be Procured

The following items and services will be acquired from external providers to support system development and deployment:

- Web hosting and server infrastructure
- OCR service or API integration tools
- Domain name and SSL security certificate
- Backup and cloud storage services
- Optional scanning equipment for digitizing documents

3.5.2 Procurement Plan Table

Item/Service	Purpose	Procurement Source	Planned Time
Web Hosting Server	Deploy the ClinicCare platform online	External Provider	Week 9
OCR API Service	Extract text from scanned medical documents	External Vendor	Week 6
Domain + SSL Certificate	Secure system access via HTTPS	External Provider	Week 10
Cloud Backup Storage	Secure archiving and disaster recovery	External Provider	Week 10
Scanner Device (Optional)	Digitizing paper medical files	Local Supplier	Week 5

Table 17: Procurement Plan for ClinicCare Project

3.6 Quality Management Plan

This section defines the quality policy, objectives, assurance, and control mechanisms adopted to ensure that the ClinicCare project deliverables meet approved requirements and stakeholder expectations.

3.6.1 Quality Policy

CLINICCARE QUALITY POLICY

The ClinicCare project is committed to delivering a high-quality Smart Medical Center Management System that meets or exceeds stakeholder expectations and project requirements. This quality policy establishes the foundation for quality planning, assurance, and control activities throughout the project lifecycle.

The ClinicCare Quality Policy is based on the following principles:

- Maintaining a strong focus on user needs, usability, and system reliability.
- Ensuring accuracy, security, and confidentiality of medical data at all times.
- Continuously improving system performance, development processes, and team collaboration.
- Promoting a quality-first culture where quality considerations guide all project decisions and activities.

All project planning, development, testing, and deployment activities for the ClinicCare system will be guided by this quality policy to ensure successful project outcomes and stakeholder satisfaction.

Figure 34: ClinicCare Quality Policy

3.6.2 Quality Objectives

QUALITY OBJECTIVES

The quality objectives for the ClinicCare project define measurable targets that ensure the delivered system meets project requirements and stakeholder expectations. The primary quality objectives include:

- **Reliability & Stability:** Achieve ≥99% system uptime during clinic working hours, and maintain a crash rate of ≤1 per week during the pilot period.
- **Performance:** Keep average page response time ≤2 seconds for 95% of user actions under normal clinic load.
- **OCR Accuracy:** Achieve ≥95% field-level accuracy for extracted key fields (e.g., name, ID, date) on a test set of at least 100 scanned documents.
- **Security & Privacy Compliance:** Ensure 100% of sensitive records are protected by role-based access control and encryption in transit (HTTPS), with 0 critical/high security findings after security testing.
- **Testing Quality:** Reach ≥90% test case pass rate for system testing before UAT, with all critical defects resolved.
- **User Acceptance:** Obtain UAT approval from clinic staff with an acceptance score of ≥80% satisfaction on the UAT evaluation form, and closure of all high-severity issues.

Figure 35: Quality Objectives for ClinicCare Project

3.6.3 Quality Assurance (Big Q)

QUALITY ASSURANCE ACTIVITIES

Quality assurance focuses on preventing defects by ensuring that project processes are properly defined, followed, and continuously improved throughout the project lifecycle. Quality assurance activities for the ClinicCare project include:

- Conducting regular requirements, design, and architecture reviews.
- Ensuring development activities follow approved system designs and technical standards.
- Performing code reviews and internal technical assessments during system development.
- Maintaining continuous coordination between the development and quality assurance teams.

Figure 36: Quality Assurance Activities for ClinicCare Project

3.6.4 Quality Control (Little q)

QUALITY CONTROL METHODS

Quality control focuses on detecting defects in project deliverables and verifying that the final system meets defined quality standards and specifications. Quality control methods for the ClinicCare project include:

- System testing and integration testing of all functional modules.
- Validation of OCR accuracy and medical data consistency.
- User acceptance testing (UAT) conducted by ClinicCare users.
- Documentation, tracking, and correction of identified defects prior to system deployment.

Figure 37: Quality Control Methods for ClinicCare Project

3.7 Risk Management Plan

Risk management is applied in the ClinicCare project to identify, analyze, and mitigate potential technical, schedule, and security risks throughout the project lifecycle.

3.7.1 Risk Management Process

- **Step 1: Risk Identification:** Identify potential risks through team brainstorming, consultation with ClinicCare staff, and review of similar healthcare system projects.
- **Step 2: Risk Analysis:** Assess each identified risk using qualitative evaluation to determine its probability of occurrence and its potential impact on the project.
- **Step 3: Risk Prioritization:** Rank risks according to likelihood and severity, allowing the project team to focus on high-priority risks.
- **Step 4: Risk Mitigation Planning:** Develop preventive actions and mitigation strategies to reduce the probability or impact of critical risks.
- **Step 5: Risk Response Implementation:** Integrate mitigation actions into the project schedule and assign responsibilities for executing risk responses.
- **Step 6: Risk Monitoring and Reporting:** Monitor risks continuously through regular project reviews and update the risk register accordingly.
- **Step 7: Risk Communication:** Communicate risk updates and mitigation progress to stakeholders to ensure transparency and alignment.
- **Step 8: Risk Review and Learning:** Conduct lessons-learned reviews after major milestones to improve future risk management practices.

3.7.2 Risk Breakdown Structure (RBS)

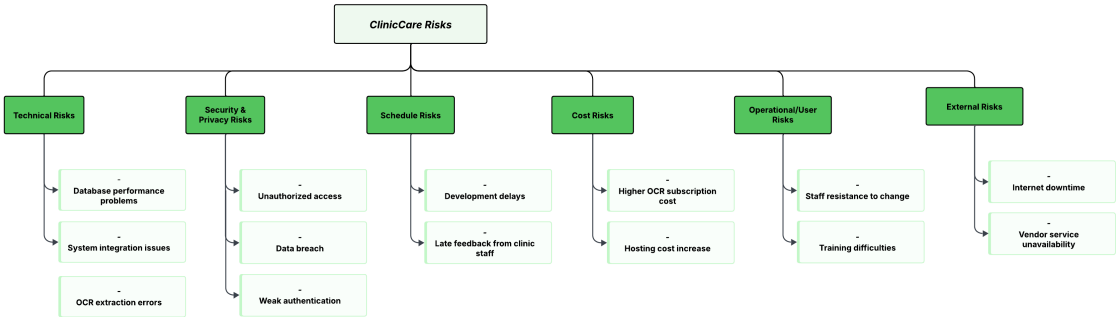

Figure 38: Risk Breakdown Structure (RBS)

3.7.3 Risk Statements Summary

RISK STATEMENTS SUMMARY

The following risk statements summarize key uncertainties that may affect the ClinicCare project during its lifecycle. These risks are derived from the Risk Breakdown Structure (RBS) and reflect the most critical potential threats to project success.

External Risks — Unpredictable:
If the hosting provider experiences unexpected downtime, then the system deployment may be delayed. If internet connectivity is unstable at the clinic, then online access to the system may be disrupted.

Internal Risks — Technical:
If the OCR module does not extract medical data accurately, then patient records may contain errors. If database performance is insufficient, then system response time may affect clinic operations.

Internal Risks — Nontechnical:
If clinic staff are not adequately trained, then system adoption may be slow. If stakeholder feedback is delayed, then project schedule milestones may be missed.

Figure 39: Risk Statements Summary for ClinicCare Project

3.7.4 Risk Register

Risk	P (1–5)	I (1–5)	Score	Owner	Response Plan	Trigger / Early Warning
OCR accuracy below target	3	4	12	Technical Lead	Improve preprocessing, expand test set, manual review for low-confidence fields, tune OCR parameters	OCR field accuracy <95% in test runs
Data privacy / access control vulnerability	2	5	10	Security Lead	RBAC enforcement, secure coding review, penetration testing, fix findings before go-live	Any high/critical security finding
Clinic staff resistance to new workflow	3	3	9	PM / Trainer	Training sessions, quick user guide, pilot support, collect feedback and iterate UI	UAT satisfaction <80% or repeated usability complaints
Delays in integration of modules	3	4	12	PM	Milestones, weekly integration tests, buffer time, defect triage	Missed milestone or rising defect count after merges
System downtime during working hours	2	4	8	DevOps / Admin	Monitoring, backups, rollback plan, performance tuning	Uptime <99% within working hours

Scoring Rule: Risk Score = Probability × Impact. Scores ≥ 12 require immediate mitigation and weekly review.

Figure 40: ClinicCare Risk Register

4 Project Monitoring and Control

Once the project management plan has been developed and project execution begins, continuous monitoring and control are required to ensure that the project remains aligned with its approved objectives. Project monitoring and control involve observing performance, measuring actual results, and taking corrective or preventive actions when deviations occur.

In the ClinicCare project, monitoring and control activities are performed throughout the execution phase to ensure that project scope, schedule, cost, quality, and risk remain within approved baselines. These activities support informed decision-making, effective risk mitigation, and successful project delivery.

4.1 Project Elements to Monitor and Control

During project execution, the project management team focuses on monitoring and controlling five key project elements:

- Scope
- Schedule
- Cost
- Quality
- Risk

Effective monitoring of these elements ensures that deviations are detected early and corrective actions are implemented before they negatively impact project objectives.

4.2 Monitoring and Controlling Scope

Scope monitoring and control in the ClinicCare project are performed to ensure that all executed work remains aligned with the approved scope statement and contractual agreements. Throughout project execution, the project management team actively monitors requested changes to prevent unauthorized expansion of project scope.

During the implementation phase, ClinicCare stakeholders may request additional features or workflow modifications. These requests are not implemented directly. Instead, they are formally reviewed by the project manager to assess their impact on project

scope, schedule, cost, and quality.

To control scope and prevent scope creep, the ClinicCare project applies structured scope monitoring and control activities supported by a formal change order procedure. This approach ensures that approved changes are incorporated in a controlled manner while maintaining alignment with project objectives.

4.3 Scope Monitoring and Control Activities

Scope monitoring and control in the ClinicCare project are carried out through three coordinated activities that collectively ensure effective control of project scope during execution. These activities define how deliverables are accepted, how changes are authorized, and how project documentation is kept accurate and up to date.

The three main scope monitoring and control activities are:

- Acceptance of deliverables
- Management of the change-order process
- Updating project documents

Figure 41 illustrates the scope monitoring and control process applied throughout the ClinicCare project lifecycle.

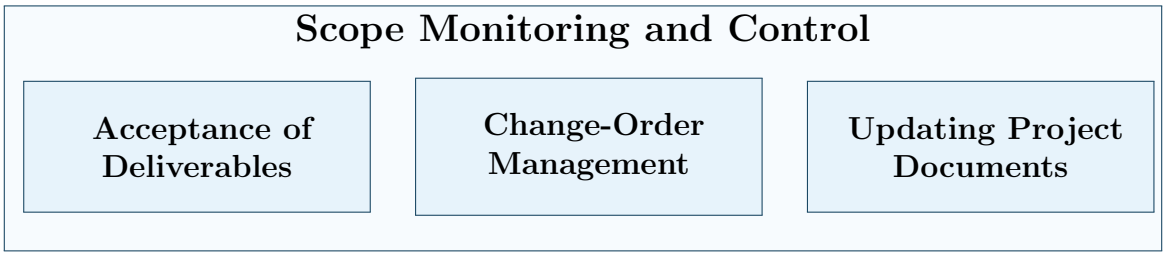

Figure 41: Scope Monitoring and Control Process Applied in the ClinicCare Project

4.3.1 Acceptance of Deliverables

Acceptance of deliverables ensures that all outputs produced during the ClinicCare project meet the predefined acceptance criteria agreed upon during the planning phase. Each deliverable is reviewed by authorized ClinicCare representatives to verify compliance with functional and quality requirements.

Formal acceptance is confirmed through customer sign-off, indicating that the deliverable

satisfies project specifications. Once accepted, deliverables cannot be modified unless a formal change order is initiated. This process protects the project team from unauthorized rework and establishes clear responsibility boundaries.

Figure 42 presents the deliverable acceptance and sign-off form used to formally approve completed deliverables in the ClinicCare project.

DELIVERABLE ACCEPTANCE FORM

Project Information

Project Name:	Smart Medical Center Management System for ClinicCare
Deliverable Name:	_____
Deliverable ID:	D-_____
Review Date:	_____
Reviewed By:	ClinicCare Representative

Acceptance Criteria Verification:

☐ All functional requirements satisfied
☐ Quality standards and performance benchmarks met

☐ Documentation completed and delivered

☐ Testing completed successfully

Acceptance Decision:

☐ **Accepted** (Ready for Deployment)	☐ **Not Accepted** (Rework Required)

Customer Signature: _____

Date: _____

Project Manager Signature: _____

Date: _____

Figure 42: Deliverable Acceptance and Sign-Off Form for ClinicCare

Figure 43 shows a sample filled deliverable acceptance form, demonstrating how deliverable approval is documented after verification.

DELIVERABLE ACCEPTANCE FORM (FILLED SAMPLE)

Project Information

Project Name:	Smart Medical Center Management System for ClinicCare
Deliverable Name:	Appointment Scheduling Module
Deliverable ID:	D-APPT-03
Review Date:	12/02/2026
Reviewed By:	ClinicCare Representative (Front Desk Supervisor)

Acceptance Criteria Verification:

■ All functional requirements satisfied (Booking, rescheduling, cancellation tested)
■ Quality standards and performance benchmarks met (Avg. response $\leq$ 2s)
■ Documentation completed and delivered (User guide + admin notes)

■ Testing completed successfully (System test pass rate: 95%)

Acceptance Decision:

■ **Accepted**	□ **Not Accepted**
(Ready for Deployment)	(Rework Required)

Customer Signature:

A. Mansour

Date: 12/02/2026

Project Manager Signature:

Hala (Project Manager)

Date: 12/02/2026

Figure 43: Sample Filled Deliverable Acceptance and Sign-Off Form for ClinicCare

4.3.2 Management of the Change-Order Process

Managing the change-order process is a critical responsibility of the project manager in the ClinicCare project. Change orders are used to formally document, evaluate, and authorize any work that falls outside the approved project scope.

Each change request is assessed for its impact on scope, schedule, cost, quality, and risk before approval. Approved change orders result in controlled updates to project baselines, while rejected requests are documented to prevent informal scope expansion.

Figure 44 illustrates the workflow followed by the project management team to evaluate, approve, and control change requests in the ClinicCare project.

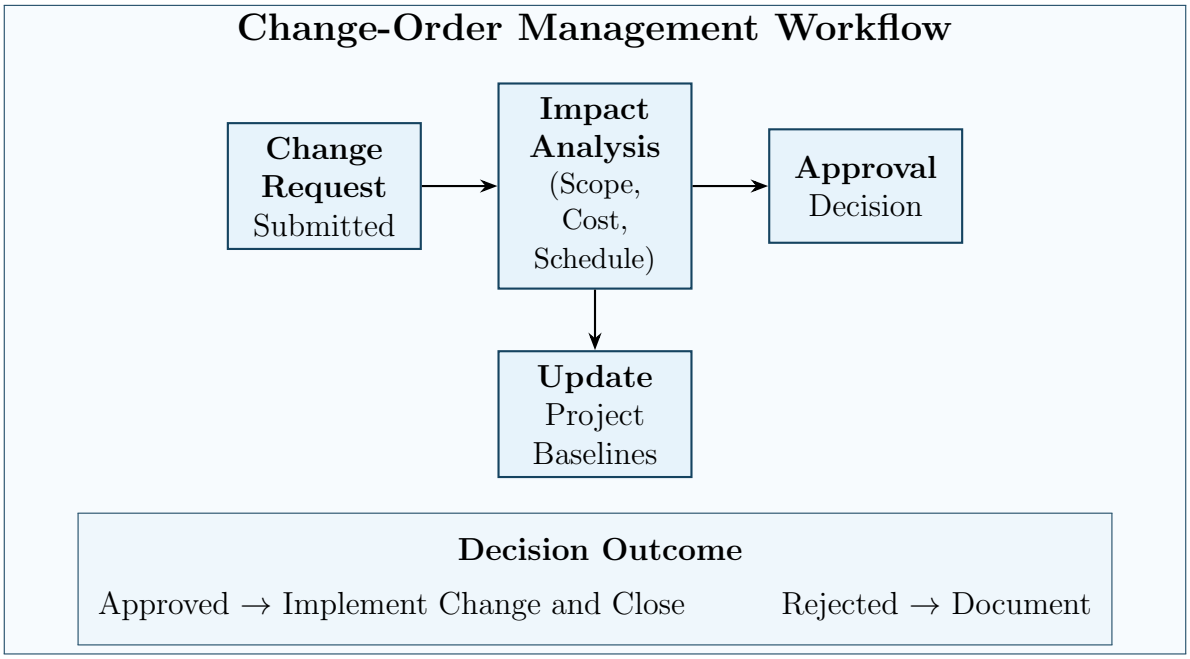

Figure 44: Change-Order Management Workflow Applied in the ClinicCare Project

4.3.3 Change Order Form (ClinicCare)

Figure 45 presents the standard change order form used to document and control scope changes in the ClinicCare project.

CHANGE ORDER FORM

Project Number: CC-PM-01
Project Name: Smart Medical Center Management System for ClinicCare
Change Order Number: CO-01
Date:

Description of the Change:

Not valid until signed by the Customer and the Project Manager

---

The Contract Sum prior to this Change Order was _____
The Contract Sum will be (increased/decreased/unchanged) _____
by
The new Contract Sum including this Change Order will be _____
The Contract Time will be (increased/decreased/unchanged) _____
by
The Date of Completion as of this Change Order is now _____

Customer	**Project Manager**
_____	_____
Date: _____	Date: _____

Figure 45: Change Order Form Template Used to Control Scope Changes in ClinicCare

Figure 46 shows a sample filled change order form, illustrating how an approved scope change is documented along with its impact on cost and schedule.

CHANGE ORDER FORM (FILLED SAMPLE)

Project Number:	CC-PM-01
Project Name:	Smart Medical Center Management System for ClinicCare
Change Order Number:	CO-02
Date:	18/02/2026

Description of the Change:

Add an SMS reminder feature for appointments, including: (i) automated SMS reminders sent 24 hours before the appointment, (ii) a configuration screen to enable/disable reminders per department, and (iii) logging of sent messages for audit purposes. This change is requested to reduce no-show rates and improve clinic scheduling efficiency.

Not valid until signed by the Customer and the Project Manager

---

The Contract Sum prior to this Change Order was	12,000 ILS
The Contract Sum will be (increased/decreased/unchanged) by	+1,200 ILS
The new Contract Sum including this Change Order will be	13,200 ILS
The Contract Time will be (increased/decreased/unchanged) by	+5 days
The Date of Completion as of this Change Order is now	05/03/2026

Customer	**Project Manager**
ClinicCare Director: M. Saleh	*Hala (Project Manager)*
Date: 18/02/2026	Date: 18/02/2026

Figure 46: Sample Filled Change Order Form Used to Control Scope Changes in ClinicCare

4.3.4 Updating Project Documents

Following the approval of any change order, all affected project documents are updated accordingly. In the ClinicCare project, this includes updates to the scope statement, project schedule, cost baseline, and risk register.

Maintaining accurate and up-to-date project documentation supports effective monitoring and control, ensures valid project closeout, and enables meaningful lessons-learned

analysis at the end of the project.

Table 18 summarizes the key project documents that are updated following the approval of a change order in the ClinicCare project.

Table 18: Project Documents Updated After Approved Change Orders

Project Document	**Update Trigger**	**Purpose of Update**
Scope Statement	Approved scope change	Reflect newly approved or removed requirements
Project Schedule	Change affecting duration	Update task timelines and milestones
Cost Baseline	Cost-related change order	Adjust budget and cost tracking
Risk Register	Change introduces new risk	Identify and mitigate emerging risks
WBS Dictionary	Scope modification	Update work package descriptions

Figure 47 illustrates how approved change orders trigger systematic updates to key project documents in the ClinicCare project.

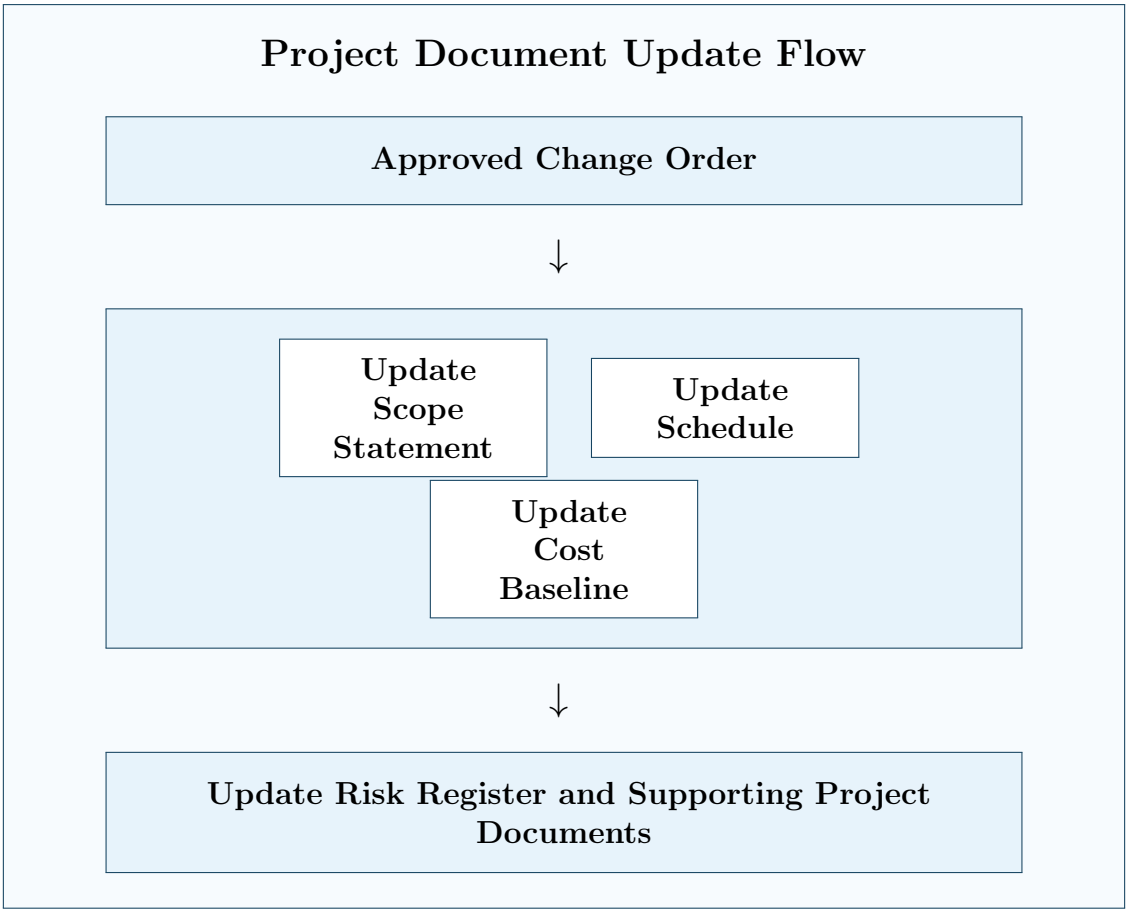

Figure 47: Document Update Process Following Approved Change Orders

4.4 Monitoring and Controlling the Schedule

Monitoring and controlling the project schedule is one of the most critical responsibilities of the project manager in the ClinicCare project. The project schedule directly influences project cost, resource allocation, and overall performance.

Throughout project execution, actual progress is continuously measured and compared against the approved baseline schedule. Any deviations are analyzed promptly to determine their causes and potential impact on project milestones and the overall completion date.

When schedule variances are identified, corrective actions are evaluated to bring the project back on track while minimizing negative effects on cost, quality, and risk. Effective schedule control supports timely decision-making, helps avoid cascading delays, and ensures that contractual deadlines are met.

Figure 48 illustrates the schedule monitoring and control cycle applied throughout the

execution of the ClinicCare project.

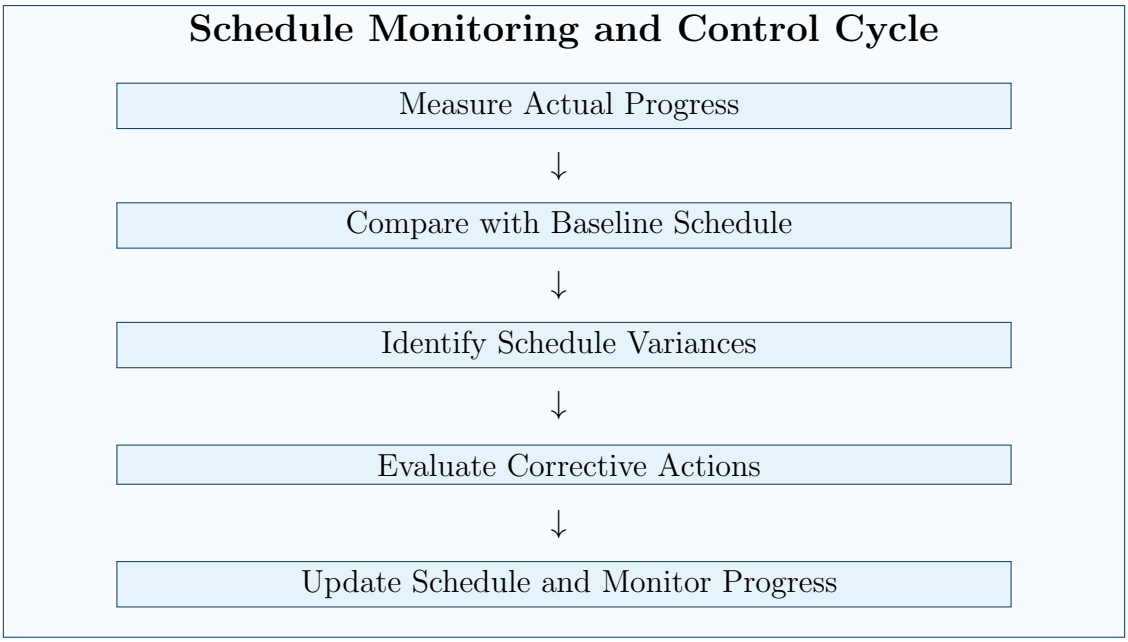

Figure 48: Schedule Monitoring and Control Cycle in the ClinicCare Project

4.5 Schedule Control Actions for Delays

When the ClinicCare project experiences schedule delays, the project manager evaluates a limited set of corrective actions. Each option has direct implications on project cost, human resources, and profitability. Schedule recovery decisions are therefore made carefully to balance time constraints with budget and quality considerations.

The primary corrective options available to the project manager include:

- Increasing project staff to accelerate delayed activities
- Approving overtime work for existing project personnel
- Subcontracting specific activities to external providers
- Accepting the risk of missing an activity deadline

Each of these options may affect other elements of the project, particularly the cost baseline and resource plan. As a result, schedule control decisions are evaluated holistically rather than in isolation.

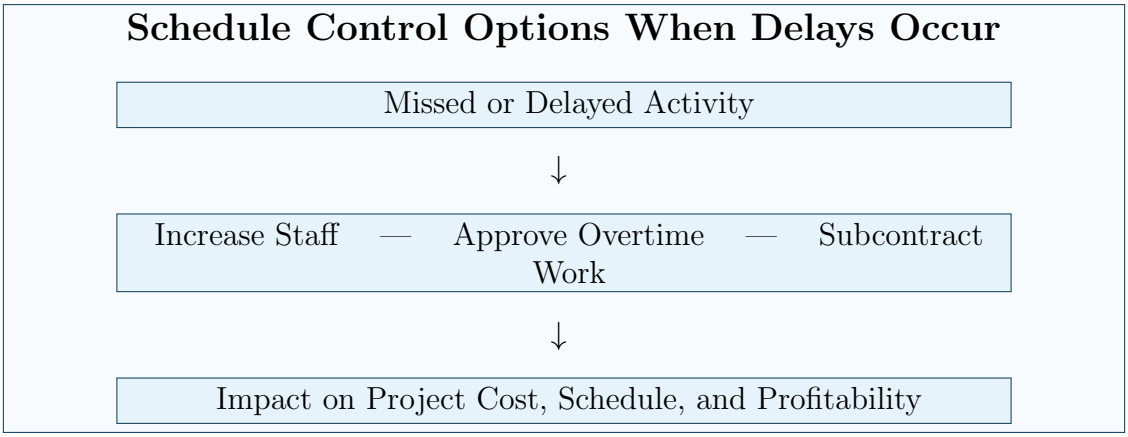

Figure 49: Schedule Control Actions Applied When the Project Falls Behind Schedule

4.6 Monitoring and Controlling Costs

Completing the ClinicCare project within the approved budget is one of the fundamental success criteria. Therefore, cost monitoring and control are applied throughout execution to ensure that actual expenditures remain aligned with the authorized cost baseline.

In the ClinicCare project, the project manager monitors costs using the cost baseline as the primary reference for tracking planned versus actual spending. Any deviations are analyzed to determine their cause, and corrective actions are planned before cost overruns escalate. Budget updates are applied only after variance causes are analyzed and the proposed adjustment is formally approved through project control procedures, including the change-order process when scope is affected.

Cost control decisions are evaluated alongside schedule and quality constraints to avoid shifting problems from one baseline to another. The project manager actively influences actions that affect the budget, ensures all changes follow formal approval processes, and maintains strict budget discipline throughout all project phases.

Figure 50 illustrates the relationship between the cost baseline, actual expenditures, and variance analysis in the ClinicCare project.

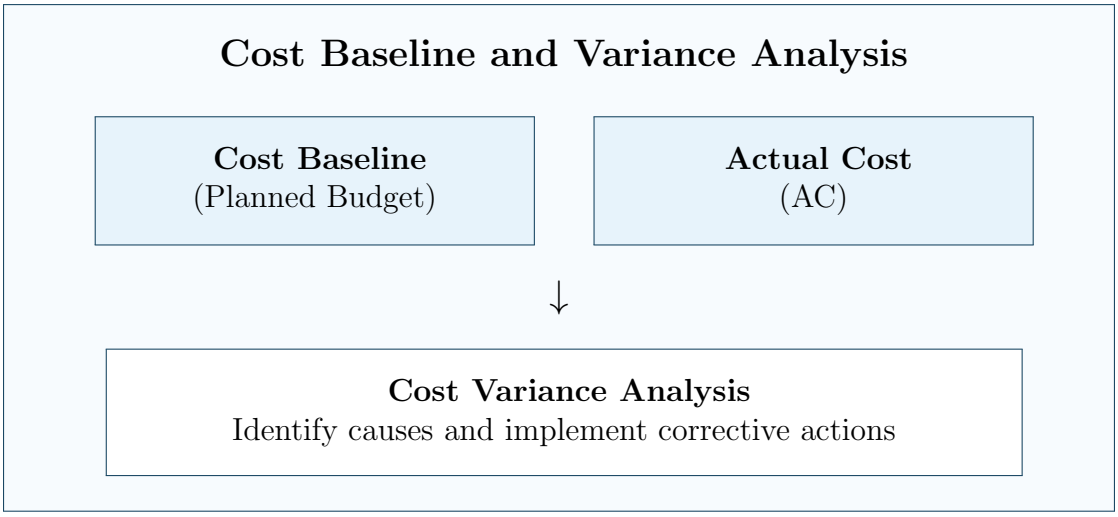

Figure 50: Cost Baseline and Variance Analysis in ClinicCare

4.6.1 Cost Monitoring and Control Responsibilities

The project manager is responsible for ensuring disciplined cost control in the ClinicCare project through the following practices:

- **Influencing factors that create changes to the cost baseline:** Monitoring work performance and acting early when performance does not meet expectations.
- **Ensuring timely action on change requests:** Processing budget-related change requests promptly to prevent small issues from becoming costly problems.
- **Managing approved changes effectively:** Ensuring that approved changes are implemented efficiently without generating additional rework.
- **Preventing budget overruns:** Tracking expenditures continuously to avoid exceeding budget limits in any reporting period.
- **Understanding cost variances:** Investigating why variances occurred and selecting appropriate corrective actions.
- **Monitoring work performance against funds expended:** Ensuring that work packages deliver expected outputs within allocated resources.
- **Preventing unapproved changes from consuming resources:** Enforcing scope control to avoid scope creep and unauthorized spending.
- **Informing stakeholders of approved cost impacts:** Communicating cost implications of approved changes before implementation.

4.6.2 Cost Monitoring and Control Methods

The ClinicCare project applies a set of complementary methods to support cost monitoring and control, as summarized in Figure 51.

COST MONITORING AND CONTROL METHODS

- **Earned Value Management (EVM):** Integrated approach to monitoring scope, cost, and schedule
- **Variance Analysis:** Comparison of planned versus actual cost performance
- **Trend Analysis:** Evaluation of cost performance trends over time to identify improving or declining patterns
- **Performance Reviews:** Regular assessment of cost performance at predefined checkpoints
- **Cost Reporting:** Weekly and monthly cost status reports to stakeholders

Figure 51: Cost Monitoring and Control Methods Used in ClinicCare

4.6.3 Earned Value Management (EVM)

Earned Value Management (EVM) is applied in the ClinicCare project to integrate scope, schedule, and cost monitoring. EVM tracks performance using three key measures: Planned Value (PV), Earned Value (EV), and Actual Cost (AC).

- **Planned Value (PV):** The budgeted cost of the work planned to be completed at a given point in time.
- **Earned Value (EV):** The budgeted value of the work actually completed at a given point in time.
- **Actual Cost (AC):** The actual cost incurred to complete the work at a given point in time.

The following performance indices are used to evaluate cost and schedule efficiency in the ClinicCare project:

PERFORMANCE INDICES

Schedule Performance Index (SPI):
SPI = EV / PV

An SPI value less than 1.0 indicates work is behind schedule.
An SPI value greater than 1.0 indicates work is ahead of schedule.

Cost Performance Index (CPI):
CPI = EV / AC

A CPI value less than 1.0 indicates costs are over budget.
A CPI value greater than 1.0 indicates costs are under budget.

Additional Variance Measures:
Cost Variance (CV) = EV - AC
Schedule Variance (SV) = EV - PV

Figure 52: Performance Indicators for EVM Monitoring in ClinicCare

These indicators are reviewed at predefined checkpoints (e.g., end of each project phase) to determine whether corrective action is needed to bring the project back within approved baselines.

4.6.4 Key EVM Terminology

The following table summarizes the key cost monitoring and control terminology used in the ClinicCare project:

Table 19: Key Cost Monitoring and Control Terminology Used in ClinicCare

Term	Meaning
BAC	Budget at Completion
PV	Planned Value (budgeted cost of work scheduled)
EV	Earned Value (budgeted cost of work performed)
AC	Actual Cost (actual cost of work performed)
SV	Schedule Variance (EV - PV)
CV	Cost Variance (EV - AC)
SPI	Schedule Performance Index (EV / PV)
CPI	Cost Performance Index (EV / AC)
EAC	Estimate at Completion
ETC	Estimate to Complete
TCPI	To-Complete Performance Index

4.6.5 EVM Application Example

At the end of Week 6, cost and schedule performance were evaluated:

- PV = 18,500 ILS
- EV = 16,200 ILS
- AC = 20,400 ILS

Performance indices: SPI = 0.88 (behind schedule), CPI = 0.79 (over budget). Corrective actions: resource reallocation, stricter change control.

4.6.6 Cost Variance Analysis and Response

Cost variance analysis compares planned cost performance to actual cost performance to identify deviations from the approved cost baseline. When variances are detected, the project manager analyzes their causes and implements appropriate corrective actions.

Table 20: Cost Variance Response Actions

Variance Type	Possible Cause	Corrective Action
Over Budget	Poor work performance	Provide support, training, or reassign tasks
Over Budget	Inaccurate cost estimates	Re-estimate remaining work packages
Over Budget	Scope creep / unapproved work	Enforce change control and document violations
Over Budget	Unexpected technical issues	Allocate contingency reserves
Under Budget	Work completed faster than planned	Accelerate dependent activities where possible
Under Budget	Underutilized resources	Optimize allocation without harming quality

4.6.7 Cost Performance Reporting

Cost performance is reported to ClinicCare stakeholders on a regular basis to ensure transparency and enable timely decision-making. Cost reports include:

- Budget versus actual expenditure summary
- Cost variance analysis (CV and CPI)

- Schedule variance analysis (SV and SPI)
- Forecast for completion costs (EAC)
- Recommended corrective actions

Figure 53 presents the continuous cycle used to monitor, analyze, and control costs during project execution.

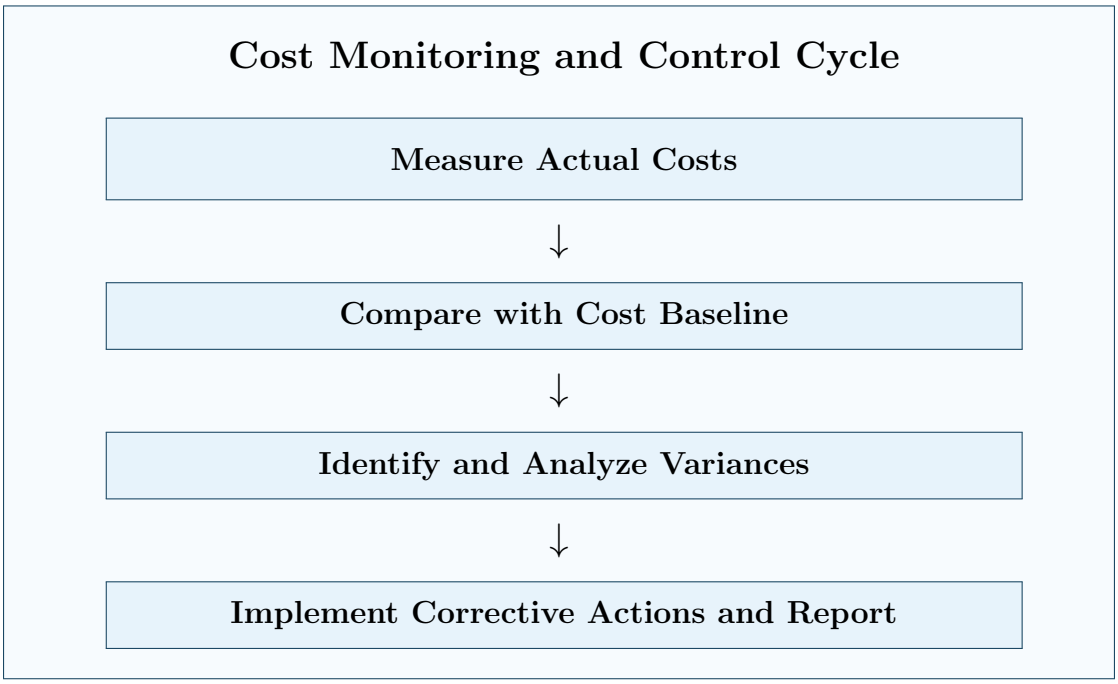

Figure 53: Cost Monitoring and Control Cycle in the ClinicCare Project

4.7 Monitoring and Controlling Quality

Quality monitoring and control in the ClinicCare project ensure that all deliverables meet or exceed the specifications defined during the planning phase. Quality activities are applied at two levels: *quality management (Big Q)* and *quality control (Little q)*.

Big Q focuses on verifying that the planned quality processes are followed consistently, while Little q focuses on day-to-day execution of quality control tools to detect and correct defects early. Both levels are essential to ensure that ClinicCare satisfies functional requirements, security constraints, and user expectations.

Figure 54 illustrates the two levels of quality monitoring and control used in ClinicCare.

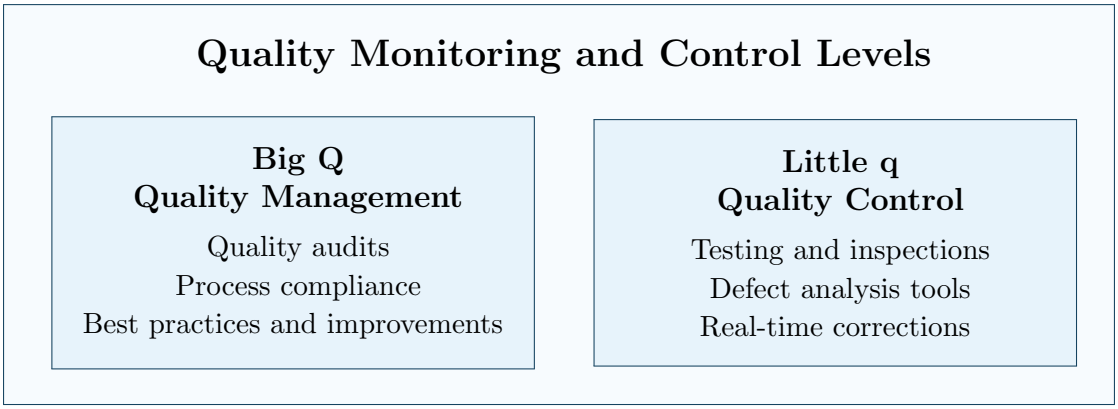

Figure 54: Two levels of quality monitoring and control applied in ClinicCare.

4.7.1 Big Q Monitoring: Quality Audits

Quality audits are conducted throughout the ClinicCare project to confirm that the quality plan is implemented as intended and that quality responsibilities are properly executed. Audits verify that:

- Quality processes are followed and documented.
- Quality tools and checklists are applied correctly.
- Quality performance reports are prepared and reviewed.
- Corrective actions are tracked until closure.

Audit results support immediate corrective action and may require updates to project documents such as the budget, schedule, and quality plan. In addition, effective practices identified during audits are documented as lessons learned for future projects.

4.7.2 Little q Monitoring: Quality Control Tools

Quality control in ClinicCare is performed continuously during development, integration, and testing. Instead of relying only on after-the-fact inspection, the project team applies quality tools to detect defects early and reduce rework cost. Figure 55 summarizes the main tools used.

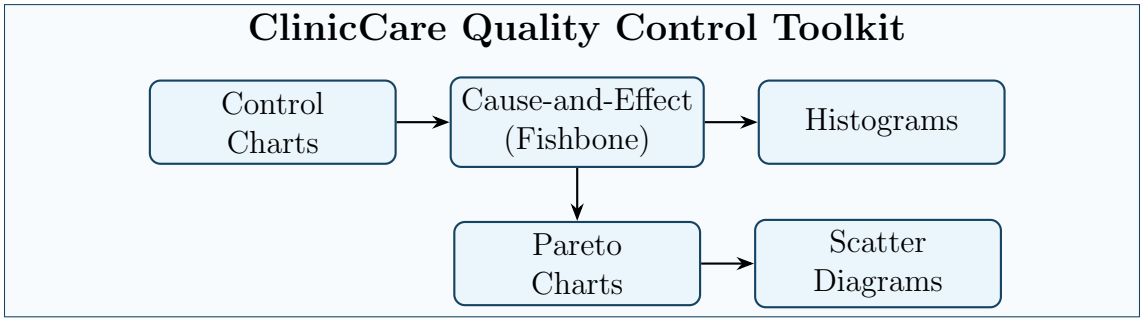

Figure 55: Core quality control tools used in ClinicCare to detect and analyze defects during execution.

4.7.3 Quality Control Process

The ClinicCare quality control process follows a systematic approach to identify defects, analyze their root causes, apply corrective actions, and verify fixes before release. Figure 56 presents the operational quality control flow.

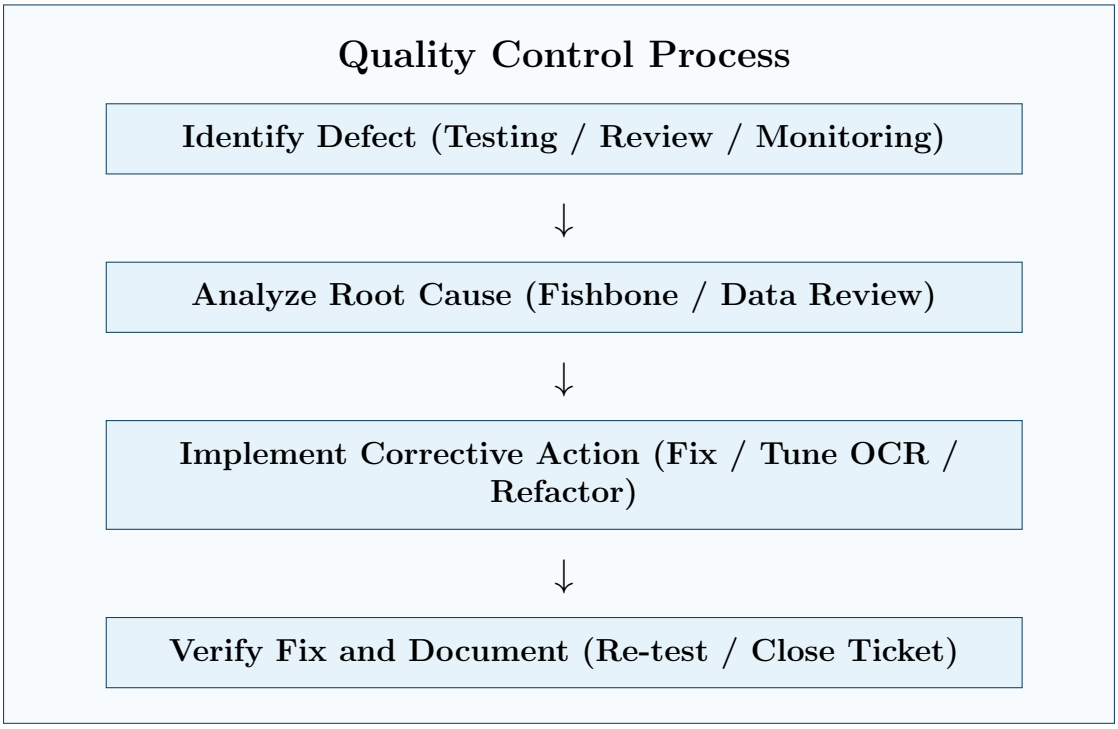

Figure 56: Operational quality control process applied in ClinicCare.

4.7.4 Defect Classification and Response

Defects identified during quality control are classified by severity to prioritize corrective actions. Table 21 summarizes the classification scheme used in ClinicCare.

Table 21: Defect Classification and Response Actions

Severity	Description	Response Action
Critical	System crash, data loss, security breach	Immediate fix required before release
High	Major functional failure affecting core features	Fix required in the current phase
Medium	Minor functional/workflow issue	Fix scheduled for current or next iteration
Low	Cosmetic/usability concern	Fix planned based on resources and priority

4.7.5 Testing and Validation Activities

ClinicCare applies multiple levels of testing to ensure comprehensive quality control, covering functional correctness, integration reliability, performance stability, and security compliance.

Table 22: Testing Levels and Objectives

Testing Level	Objective	Responsible Party
Unit Testing	Verify correctness of individual modules	Development Team
Integration Testing	Verify correct interaction among modules	Development Team
System Testing	Verify system meets requirements end-to-end	QA Team
User Acceptance Testing	Verify usability and workflow fit	ClinicCare Users
Performance Testing	Validate response time and stability under load	QA Team
Security Testing	Validate access control and data protection	IT Support & QA Team

4.7.6 Quality Control Charts (Templates)

To support real-time quality monitoring, ClinicCare uses chart-based tools to identify instability, frequent defect sources, and correlations between variables. The following figures provide example templates that can be populated using project measurements.

Control Charts

Control charts are used in ClinicCare to monitor process stability during repeated and measurable activities such as defect rates across sprints, OCR field accuracy over test batches, average response time during peak clinic hours, or document-processing completion time. The objective is to ensure that the selected quality metric remains within acceptable control limits (UCL and LCL). If values exceed limits or show non-random patterns, the process is investigated and corrective actions are triggered (e.g., bug fixing, configuration tuning, or workflow adjustment).

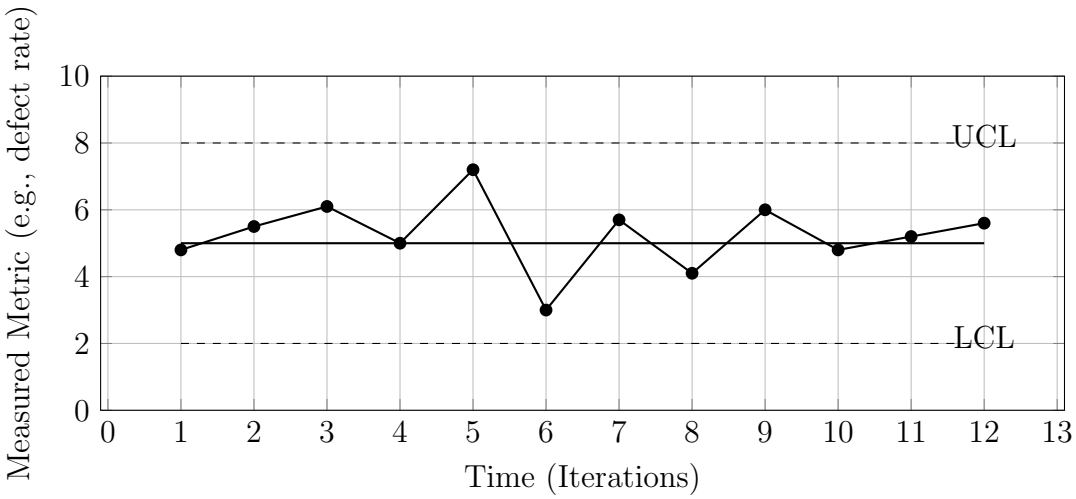

Figure 57: Example control chart used in ClinicCare to track a quality metric within control limits.

Cause-and-Effect (Fishbone) Diagrams

When a quality issue is observed in ClinicCare (e.g., repeated OCR extraction failures, increasing data-entry errors, slow system response, or frequent UI/validation issues), the team applies a cause-and-effect (fishbone) diagram to identify root causes before implementing a permanent fix. Causes are analyzed across process methods, measurements, data/materials, people, tools/infrastructure, and environment.

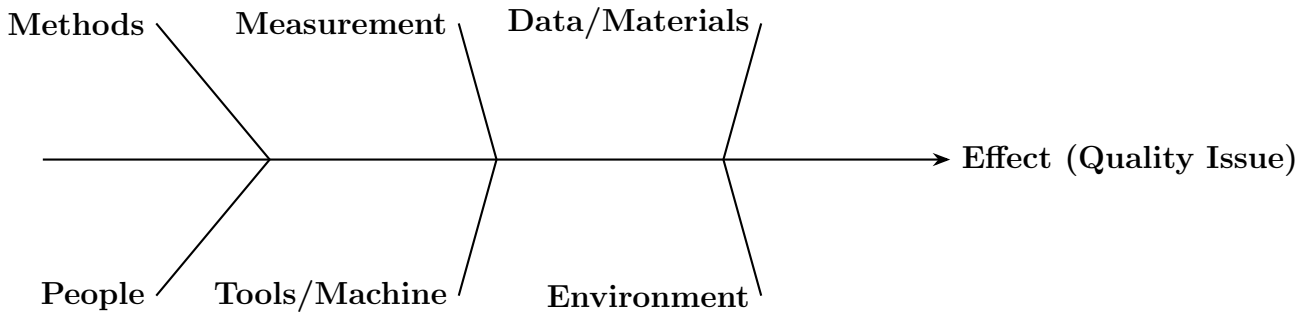

Figure 58: Fishbone template used in ClinicCare to identify root causes of quality issues.

Histograms

Histograms are used in ClinicCare to summarize defect occurrences by category over a defined period (e.g., sprint or testing cycle). This helps the team identify the most frequent issue types (e.g., data validation errors, UI bugs, integration issues, OCR formatting failures) and prioritize corrective actions accordingly.

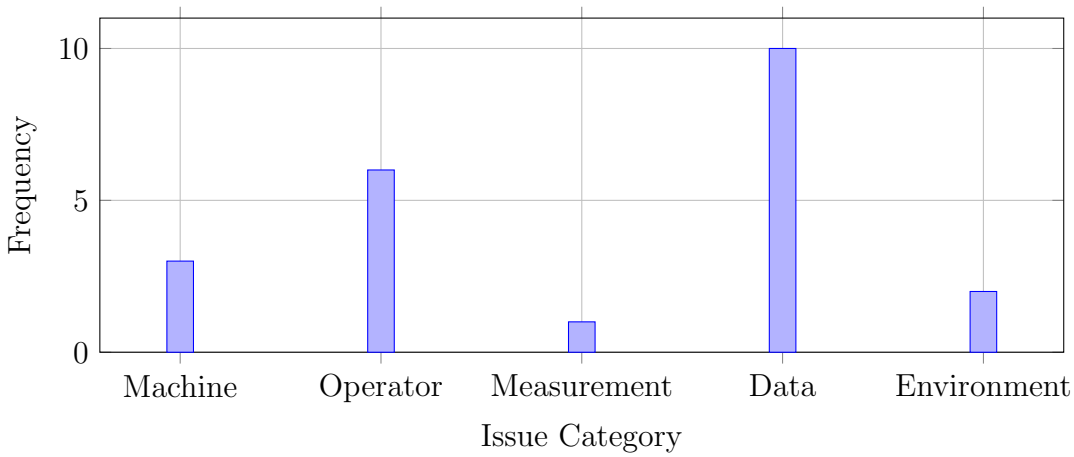

Figure 59: Histogram template summarizing issues by category within a monitoring period.

Pareto Charts

Pareto charts are applied in ClinicCare to rank defect sources from most to least frequent and highlight the small number of causes responsible for most quality issues. This supports efficient corrective action planning by focusing first on the highest-impact defect categories (e.g., data issues, UI issues, integration failures, OCR extraction defects).

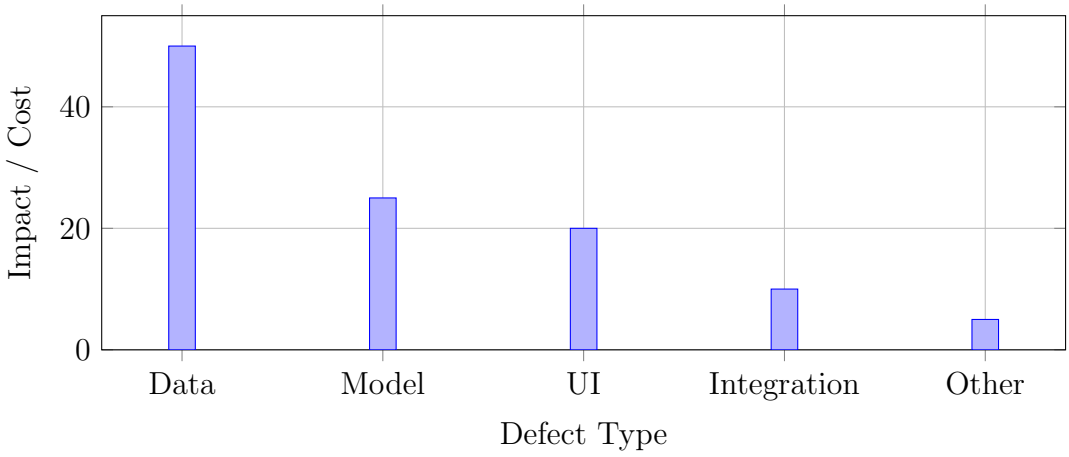

Figure 60: Pareto template used to prioritize defect sources by impact in ClinicCare.

Scatter Diagrams

Scatter diagrams are used in ClinicCare to study the relationship between two measurable variables, such as scanned image resolution versus OCR accuracy, document size versus processing time, concurrent users versus average response time, or network latency versus form submission time. This helps determine whether changes in one factor are associated with changes in another and guides improvement actions.

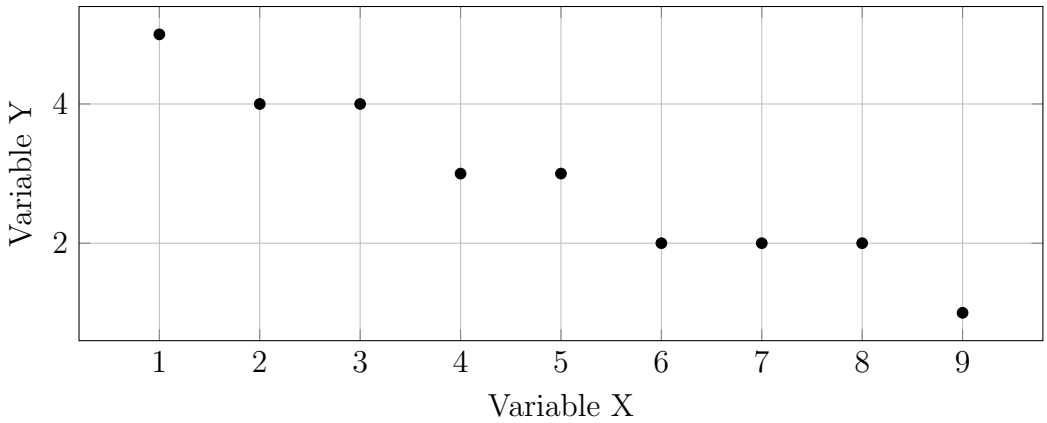

Figure 61: Scatter template to study correlation between two ClinicCare quality-related variables.

4.7.7 Quality Performance Reporting and Continuous Improvement

Quality performance is reported regularly to ClinicCare stakeholders to support transparency and timely decision-making. Reports include defect counts by severity, resolution status, test pass rates, audit findings, and corrective actions.

Lessons learned from audits and quality control activities are used to continuously im-

prove the quality plan, refine metrics, enhance team practices, and reduce recurring defects in subsequent iterations.

4.8 Monitoring and Controlling Risk

Monitoring and controlling risk in the ClinicCare project aim to ensure that project risks are continuously identified, evaluated, and managed throughout the execution phase. While risk identification and response planning are conducted during the planning phase, risk conditions may change as the project progresses. Therefore, risk monitoring is treated as an ongoing management activity rather than a one-time exercise.

During project execution, the project management team tracks previously identified risks, monitors newly emerging risks, evaluates changes in risk probability and impact, and assesses the effectiveness of implemented risk response strategies. This approach ensures that potential threats to project objectives are detected early and addressed proactively.

Risk monitoring and control activities also support informed decisions regarding the use of contingency reserves related to cost and schedule. When risk-related assumptions made during planning prove inaccurate, corrective actions are taken to prevent negative impacts on project performance.

4.8.1 Risk Evaluation During Execution

As part of continuous risk monitoring, the project manager evaluates several key risk-related questions throughout the execution of the ClinicCare project:

- Are the risk-related assumptions made during the planning phase still valid?
- Have any identified risks increased or decreased in probability or impact?
- Are there risks that are no longer relevant or have been eliminated?
- Is it necessary to activate budget contingency reserves to address new or changed risks?
- Is it necessary to activate schedule contingency reserves to mitigate emerging delays?

Answers to these questions guide the selection of corrective actions, including revising existing risk responses, developing new mitigation strategies, or applying approved contingencies.

4.8.2 Risk Monitoring and Control Process

Risk monitoring and control in the ClinicCare project follow a structured process that ensures systematic tracking, evaluation, and response to project risks. Figure 62 illustrates the process applied throughout project execution.

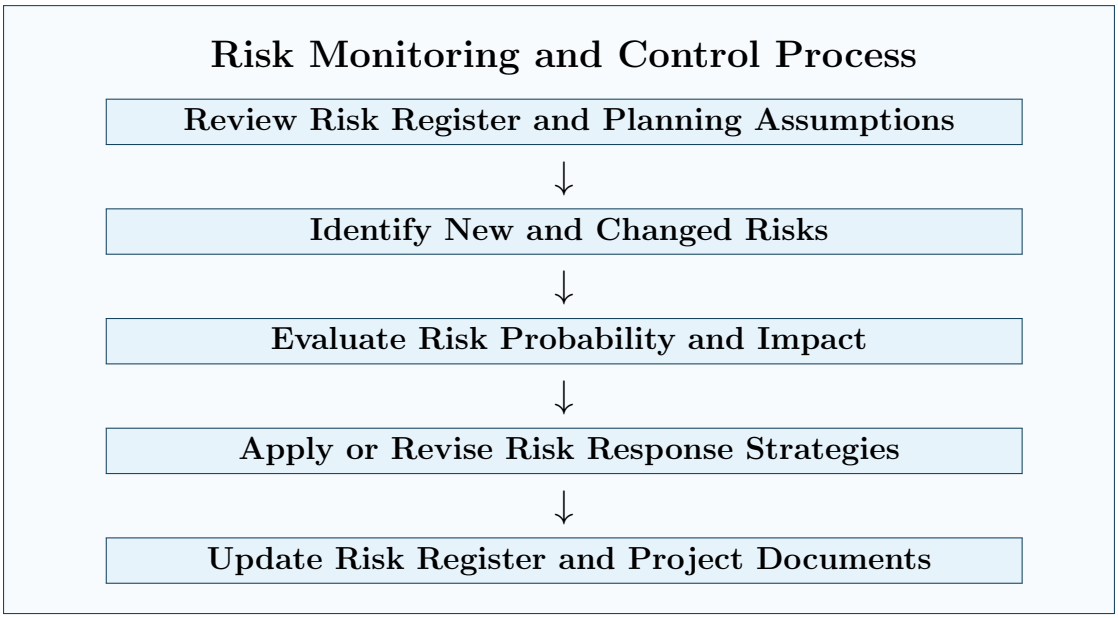

Figure 62: Risk monitoring and control process applied during the ClinicCare project execution.

4.8.3 Risk Monitoring and Control Tools

To ensure effective risk control, the ClinicCare project applies the risk monitoring tools recommended by the Project Management Institute (PMI). These tools support proactive risk identification, assessment, and response throughout project execution.

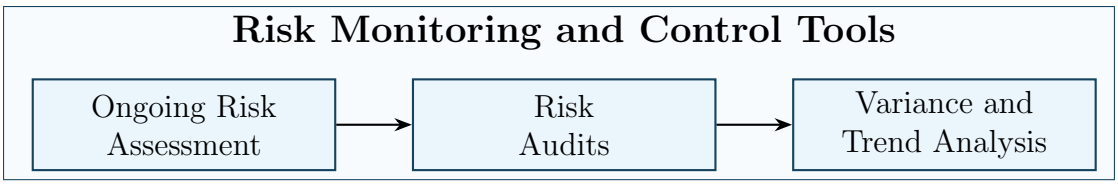

Figure 63: PMI-recommended tools for monitoring and controlling risk in ClinicCare.

- **Ongoing Risk Assessment:** Continuous evaluation of identified risks and detection of new risks introduced during execution.
- **Risk Audits:** Periodic reviews used to assess whether implemented risk response strategies are effective and properly executed.
- **Variance and Trend Analysis:** Comparison of actual project performance against risk baselines to identify unfavorable trends that may indicate emerging risks.

4.8.4 Risk Monitoring and Control Cycle

Risk monitoring in the ClinicCare project is performed as a continuous cycle to ensure timely detection and response to evolving risk conditions. Figure 64 presents the recurring risk monitoring and control cycle applied during project execution.

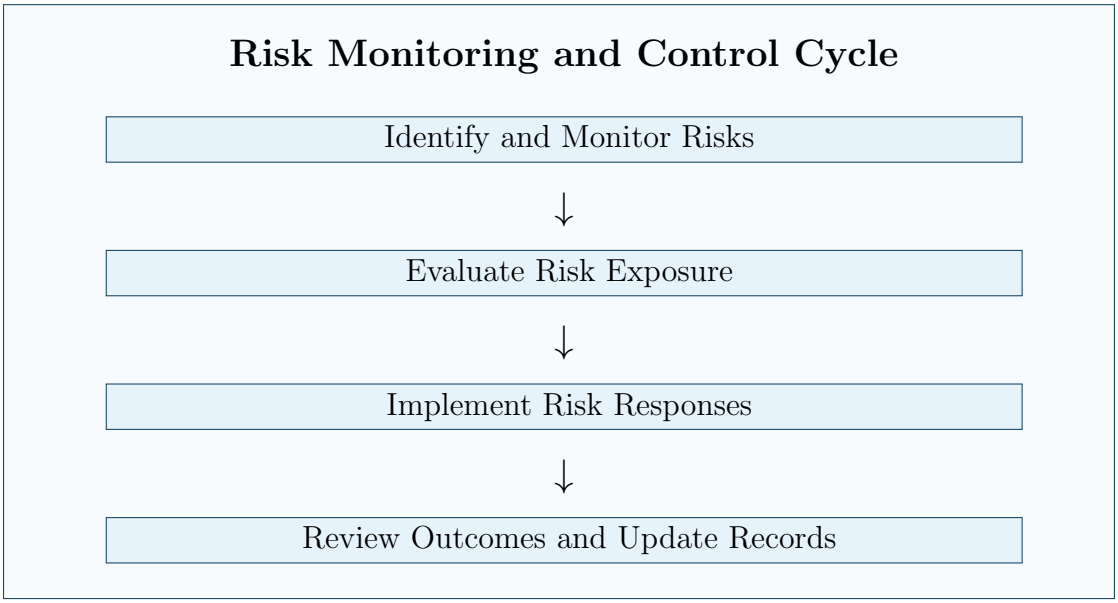

Figure 64: Continuous risk monitoring and control cycle applied in the ClinicCare project.

4.8.5 Risk Documentation and Lessons Learned

All significant findings resulting from risk monitoring and control activities are documented to support project transparency and organizational learning. The risk register is updated regularly to reflect changes in risk status, newly identified risks, revised mitigation strategies, and contingency usage.

Lessons learned related to faulty risk assumptions, emerging risks, and effective response strategies are recorded during project execution and formally reviewed during project closeout. These lessons contribute to improving risk management practices in future ClinicCare projects.

4.9 Integration of Monitoring Activities

Effective project control requires integrated monitoring of all elements. In ClinicCare:

- Scope changes affect schedule, cost, and risk
- Schedule delays may require additional resources (affecting cost)
- Quality issues require rework (affecting schedule and cost)

The project manager coordinates monitoring to ensure:

- Holistic performance assessment
- Balanced decision-making across constraints
- Timely identification of cascading impacts
- Effective stakeholder communication

4.10 Weekly Status Report (Monitoring)

Table 23: Weekly Status Report

Item	Details
Week # / Date Range	Week 3 (Mar 10–Mar 16, 2026)
Overall Status	**Yellow** (minor delay + cost pressure)
Completed This Week	- Implemented login + basic RBAC roles (Admin/Staff) - Designed OCR upload screen and validation messages - Finished database entities for Patients and Visits
Planned Next Week	- Integrate OCR API (initial pipeline + confidence score) - Implement patient registration workflow + search - Start unit tests for authentication and patient module
Key Risks This Week	- OCR accuracy may drop on low-quality scans (needs dataset + tuning) - Scope creep from new feature requests
Issues / Blockers	- Clinic sample forms/dataset delivery is late - OCR library integration docs unclear (needs workaround)
Decisions Needed	- Approve final UI layout for registration screens - Decide OCR approach: cloud OCR vs local model

4.11 Schedule Tracking (Planned vs Actual)

Table 24: Schedule Tracking Table

Task / Deliverable	Planned Start	Planned End	Actual Start	Actual End	Variance
Requirements & Scope	Mar 1	Mar 7	Mar 1	Mar 8	+1 day
UI Prototype	Mar 8	Mar 14	Mar 9	Mar 16	+2 days
RBAC & Authentication	Mar 15	Mar 21	Mar 15	Mar 20	-1 day
OCR Integration (Phase 1)	Mar 15	Mar 21	Mar 17	–	In progress (+2 days)
Patient Registration Flow	Mar 22	Mar 28	–	–	Not started

4.12 Cost Monitoring and Earned Value (EVM)

Table 25: EVM Summary (Week 3 Snapshot)

Metric	Value (ILS)	Meaning
BAC (Total Budget)	118,580	Total approved project budget
Planned Progress	25%	Week 3 out of 12 planned timeline
Actual Progress (assumed)	22%	Work completed based on deliverables status
PV (Planned Value)	29,645	BAC × planned progress (0.25)
EV (Earned Value)	26,088	BAC × earned progress (0.22)
AC (Actual Cost)	32,000	Actual spending to date
CV = EV - AC	-5,912	Negative = over budget
SV = EV - PV	-3,557	Negative = behind schedule
CPI = EV/AC	0.82	< 1 means inefficient spending
SPI = EV/PV	0.88	< 1 means behind schedule

Interpretation:

- CPI $< 1 \Rightarrow$ spending is higher than the value earned (over budget).
- SPI $< 1 \Rightarrow$ progress is slightly behind the planned schedule.
- Corrective action: freeze scope changes for 1 week, focus on OCR integration + dataset collection, reduce rework by adding quick validation tests.

4.13 Issue and Action Log

Table 26: Issue and Action Log

ID	Issue	Owner	Priority	Due Date	Status
I-01	OCR accuracy drops on low-quality scans	OCR Team	High	Mar 18	Open
I-02	Clinic sample forms/dataset delayed	PM Team	High	Mar 16	In progress
I-03	UI feedback requires minor redesign	UI Team	Medium	Mar 20	Open
I-04	Cost overrun due to extra dev hours	PM Team	Medium	Mar 21	Open

4.14 Change Request Log

Table 27: Change Request Log

CR#	Change Description	Requested By	Impact	Decision	Status
CR-01	Add appointment reminders feature	Client	Scope/Time	Approved (+1 week if included)	Approved
CR-02	Replace OCR library/tool	Team	Cost/Risk	Rejected (keep Phase 1 plan)	Closed
CR-03	Add extra user role (Reception-ist)	Client	Scope	Approved (no schedule change)	Approved